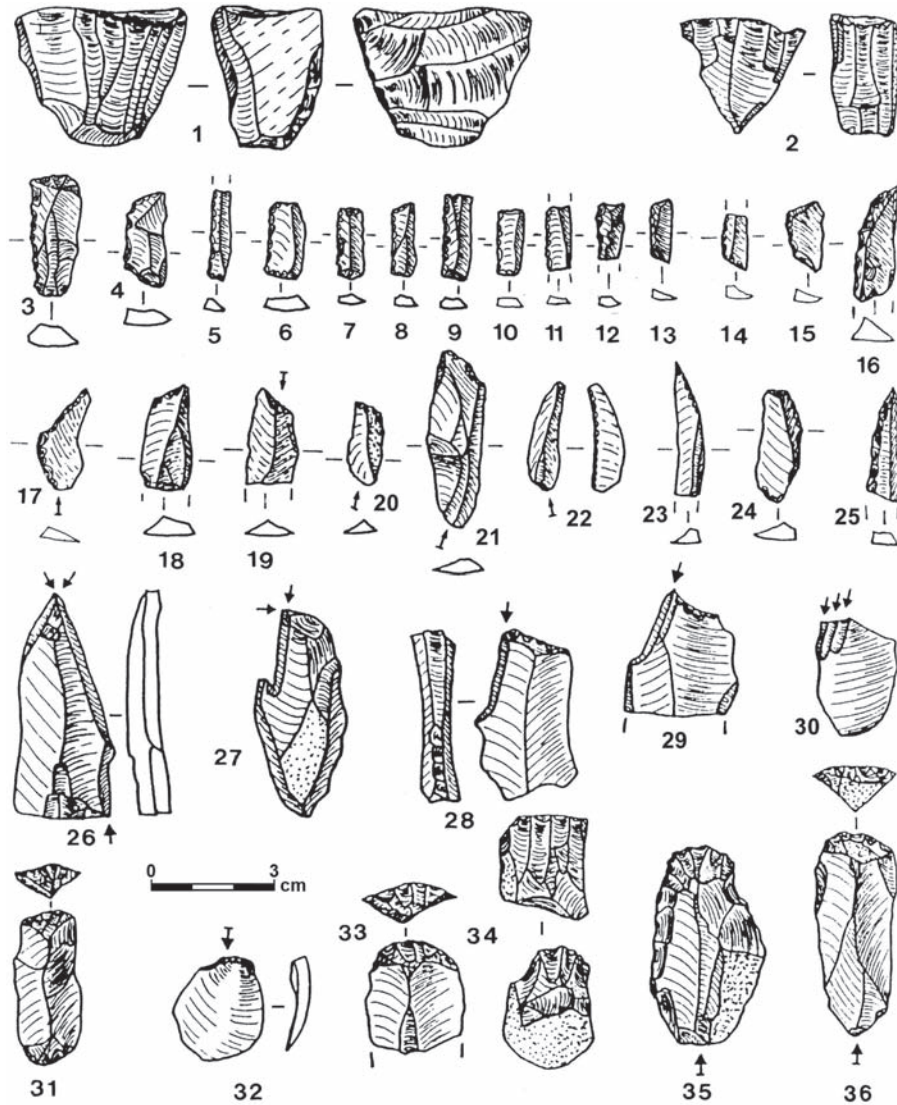




570. Khirbet Malih (C): 1-2. Bladelet core; 6-7, 9-10, 12. Rectangle; 4, 8, 13. Trapeze; 15. Asymmetric trapeze; 5, 11, 14. Rectangle/ Trapeze (fragments); 16, 24. Arch-backed bladelet; 17-21. Obliquely truncated bladelet; 22. Ventrally retouched bladelet; 23. Partially retouched bladelet; 25. Falita point (fragment); 26. Double dihedral burin; 27. Angled dihedral burin; 28-29. Burin on concave truncation; 30. Flat faced burin; 31. Double end scraper; 32. Thumbnail scraper; 33. End scraper on snapped blade; 34. Nucleiform (core) scraper; 35-36. Carinated end scraper.

Site no. 99

Qta'at el-Khalifeh

Isr. Grid Ref. 1912/1913

The flint assemblage:

Chalcolithic Period – Reasonable identification.

Artifacts without retouch and waste:

Flake 1; Blade (fragment) 1. Total – 2.

Tools and retouched artifacts:

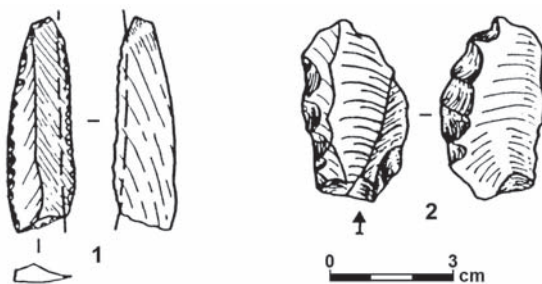
Backed and bi-truncated sickle blade 1; Bilateral denticulate 1.

Total – 2.

Total of flint artifacts – 4.

Discussion:

The assemblage is very small but the presence of the backed and bi-truncated sickle blade (Lee 1973, 250; Type E – Gopher 1989a, 95) points to the Chalcolithic Period. This is in accordance with the pottery findings attributed entirely to this period.



571. Qta'at el-Khalifeh: 1. Sickle blade; 2. Bilateral denticulate.



Site no. 134

Ro'i (1)

Isr. Grid Ref. 1964/1846

The flint assemblage:

Chalcolithic Period – possible identification.

Artifacts without retouch and waste:

Primary flake 2; Flake 4. Total – 6.

Tools and retouched artifacts:

End scraper 1; Double end scraper 1; Side scraper (fragment) 1; Fan

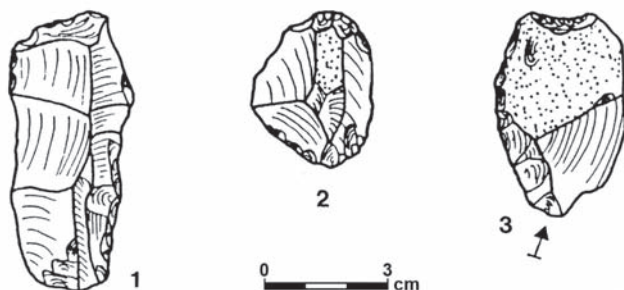
scraper (fragment)? 1; Notch 3; Retouched flake 2; Retouched blade 2; Backed knife 1. Total – 12.

Total of flint artifacts – 18.

Discussion:

Despite the fact that the assemblage does not contain specific Chalco-

lithic artifacts, no artifacts indicating a different period were present. The fragment of the fan scraper, the backed knife and the pottery findings support the assumption that the site should be attributed to the Chalcolithic Period.



572. 1. Backed knife; 2. Double end scraper; 3. End notch.



Site no. 144

Khirbet Umm Kharaz

Isr. Grid Ref. 1980/1781

See Ras el-Kharube sites – Kh. I below.



Site no. 145

Abu Loz

Isr. Grid Ref. 1841/1881

The flint assemblage:

Lower Palaeolithic Period ? — possible identification (a single artifact). The other artifact was not

defined.

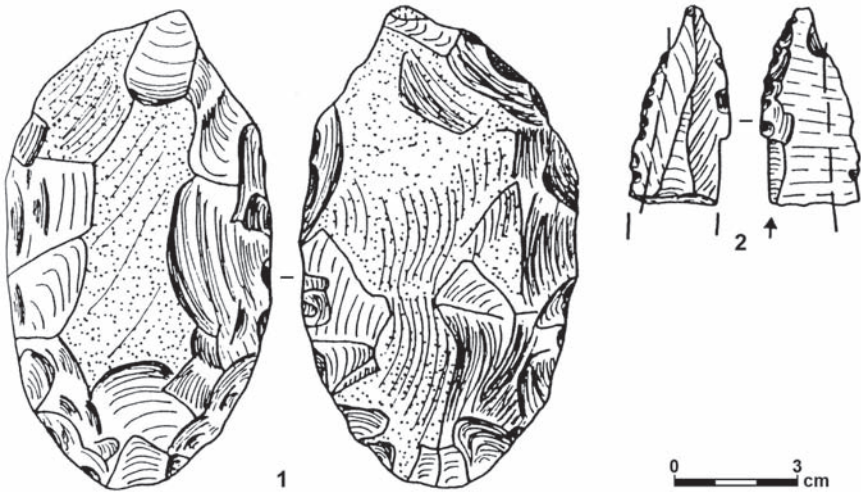
Tools and retouched artifacts:

Sickle blade (fragment) 1; Bifacial hand axe 1. Total – 2.

Discussion:

Two artifacts are not sufficient in order to define a site. The broken sickle blade could be the tip of a Neolithic reaping knife or of a Cananean blade with a sporadic burin scar. The Ovoid hand axe (Bordes 1961) was produced from a flint cobble. Such hand axes are 27% of the Ma'ayan Barukh assemblage from the Lower Palaeolithic Period (Stekelis and Gilead 1966,

15). Similarly to our specimen the cortex on many of these artifacts remained preserved (Gilead 1970). To some extent similar tools reappear during the Neolithic (Olami 1984, 139, 151-152), but most of these tools show signs of battering and use wear on the circumference. In face of the fact that no further evidence for the Lower Palaeolithic Period in the studied area was reported, the attribution of the tool to this period is uncertain.



573. Abu Loz: 1. Ovoid hand axe ?, Axe? 2. Sickle blade (fragment).

Site no. 169

Khirbet 'Ain Farr

Isr. Grid Ref. 1855/1829

The flint assemblage:

Chalcolithic Period – possible identification.

Early Bronze Age – highly reasonable identification.

Artifacts without retouch and waste:

Flake 5; Blade 1; Bladelet (fragment) 2; Chip 1. Total – 9.

Tools and retouched artifacts:

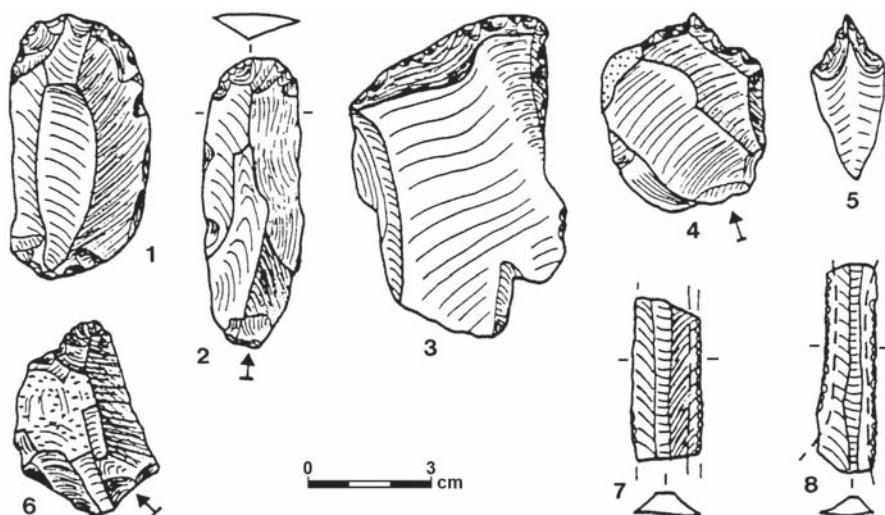
Cananean sickle blade 5; Cananean blade (fragment) 1; End scraper on flake 3; End scraper on blade 2; Double end scraper on flake 1; Shouldered end scraper on flake 1; Converging scraper on flake 1; Notch 2; Denticulate 1; Awl 1;

Borer 1; Retouched flake 5; Retouched blade 1; Core with secondary retouch 1. Total – 26.

Total of flint artifacts – 35.

Discussion:

The Cananean blades and sickle blades are *type fossils* of the Early Bronze Age (and Middle Bronze Age I as well; Rosen 1989, 214-215). Thirty percent of the pottery findings at the site were attributed to the Early Bronze Age and also a part of the flint assemblage is similarly attributed. About half of the pottery was attributed to the Chalcolithic but no prominent tools which could be attributed de-



574. Khirbet 'Ain Farr: 1. Double end scraper; 2. End scraper; 3. Converging scraper; 4. Awl; 5. Borer; 6. Shouldered end scraper; 7-8. Cananean sickle blade.

finitively to this period were found (such as adzes, axes and characteristic sickle blades). The scrapers, the notches and the borer could be

from any of these periods. It is quite difficult to find any differences between *ad hoc* tools of both periods.



Site no. 178

Jelamet el-Ahmar (A)

Isr. Grid Ref. 1898/1814

The flint assemblage:

Pottery Neolithic Period – highly reasonable identification.

Early Bronze Age – highly reasonable identification.

Artifacts without retouch and waste:

Flake core 1; Primary flake 13; Flake 11; Blade 4; Blade (fragment) 4; Chunk 9; Chip 2. Total – 44.

Tools and retouched artifacts:

Arrowhead, type Byblos (pressure flaking) 1; Arrowhead, type Amuq 1; Backed and bi-truncated sickle blade 2; Cananean sickle blade 6; Bifacial tool (fragment) 1; End scraper on flake 2; Nucleiform (core) scraper 1; Side scraper 1; Dihedral burin on flake 1; Flat faced burin on flake 1; Angled burin on primary flake 1; Tanged dihedral burin 1; Backed and truncated bladelet 4; Notch 8; Denticulate 2; Awl 3; Borer (fragment) 1; Retouched flake 4; Retouched blade 7; Cananean blade (fragment) 2; Truncation 4. Total – 54.

Total of flint artifacts – 98.

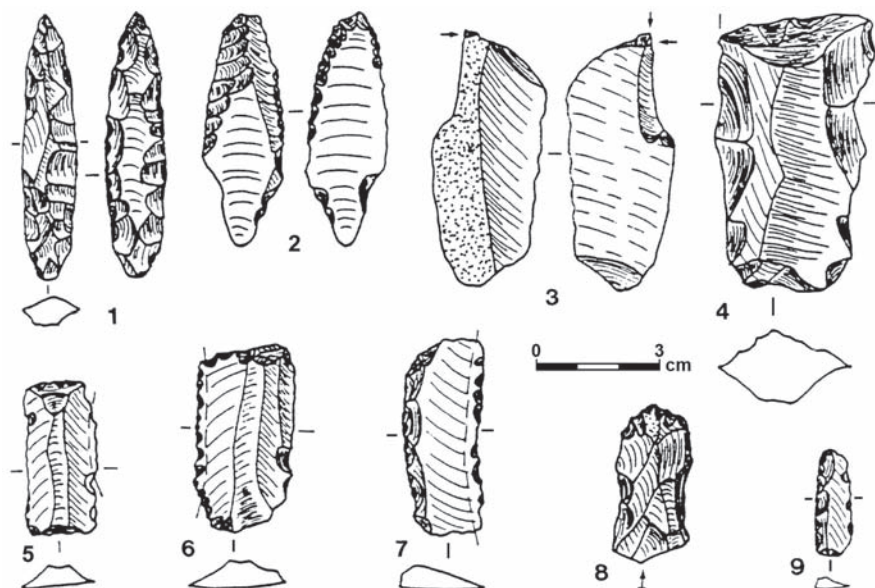
Discussion:

The Cananean sickle blade is a prominent marker of the Early Bronze Age. Backed and truncated sickle blades, and in the Negev retouched bladelets, as well, still prevailed to some extent during this period (Schick 1978; Rosen 1989). Apparently a part of the flint assemblage has to be attributed to the Early Bronze Age, similarly to the main part of the pottery findings.

Arrowheads of Byblos and Amuq types are attributed mainly to the late Pre-Pottery Neolithic B Period or early Pottery Neolithic Period (Stekelis 1966; Gopher 1981, 1985). The partial pressure flaking on the Byblos point serves as a further chronological indication for the period. The presence of burins during the Pottery Neolithic Period as documented at Sha'ar Hagolan (Stekelis 1966), seems to confirm that the other part of the assemblage originates from this period. The pottery findings from this pe-

riod at the site consolidate this dating. The absence of typical axes and sickle blades does not contradict

this assumption (see also site 179 – Jelamet el-Ahmar (B)).



575. Jelamet el-Ahmar (A): 1. Arrowhead type Amuq; 2. Arrowhead type Byblos; 3. Angled burin; 4. Bi-facial tool (fragment); 5-6. Cananean sickle blade; 7. Backed and bi-truncated sickle blade; 8. End scraper; 9. Backed and truncated bladelet.



Site no. 179

Jelamet el-Ahmar (B)

Isr. Grid Ref. 1893/1815

The flint assemblage:

Pottery Neolithic Period (main part of the assemblage) – highly reasonable identification.

Chalcolithic Period (part of the assemblage) – reasonable identification.

Early Bronze Age (part of the as-

semblage) – reasonable identification.

Artifacts without retouch and waste:

Flake core 7; Primary flake 4; Flake 14; Blade 9; Blade (fragment) 5; Chunk 7; Spherical hammerstone 4. Total – 50.

Tools and retouched artifacts:

Arrowhead, type Byblos 1; Arrowhead, type Amuq (pressure flaking) 1; Sickle blade with coarse bifacial denticulation 2; Sickle blade with coarse bifacial denticulation (fragment) 1; Truncated Cananean sickle blade 1; Thin axe with lentoid cross-section 2; Axe with polished cutting edge 2; Coarse massive axe 1; Axe (fragment) 1; Unfinished bifacial tool 1; End scraper 2; Nucleiform (core) scraper 1; Scraper on polished axe fragment 1; Side scraper 1; Dihedral burin on blade 2; Angled dihedral burin on flake 3; Notch 10; Double Notch 1; Denticulate on primary flake 2; Awl 3; Retouched flake 4; Retouched blade 2. Total – 46.

Total of flint artifacts – 96.

The stone tool assemblage:

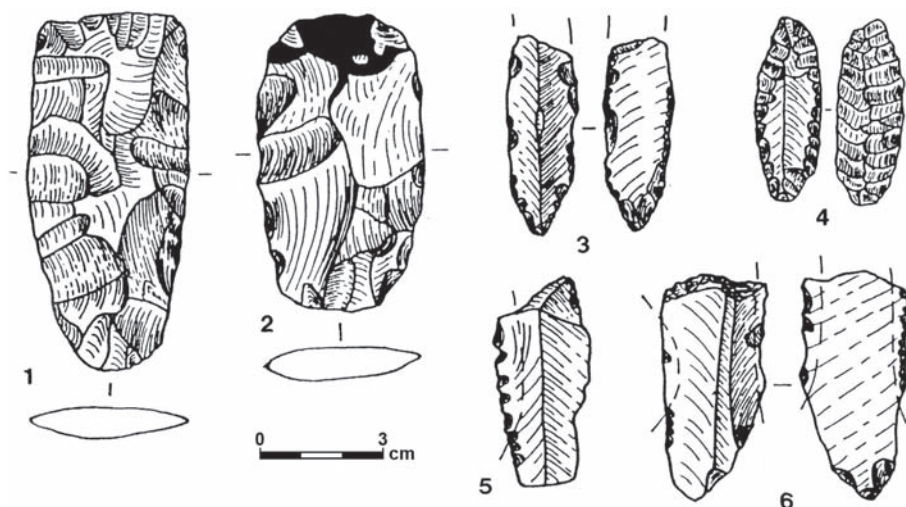
Basalt pebble (similar to a female sex organ) 1; Sandstone shaft sharpener 1; Small cup mark on a limestone pebble 1; Coarse limestone adze (hoe?) 2. Total – 5.

Discussion:

The Cananean sickle blade and probably some other tools should be attributed to the Early Bronze Age, similarly to 40% of the pottery assemblage.

No *type fossils* from the Chalcolithic Period (30% of the pottery) were found, but apparently the coarse adzes (or hoes) and some of the scrapers and notches could be attributed to this period.

Most flint and stone artifacts should be attributed to the Neolithic Period. Arrowheads of Byblos and Amuq types with pressure flaking and thin axes with a lentoid



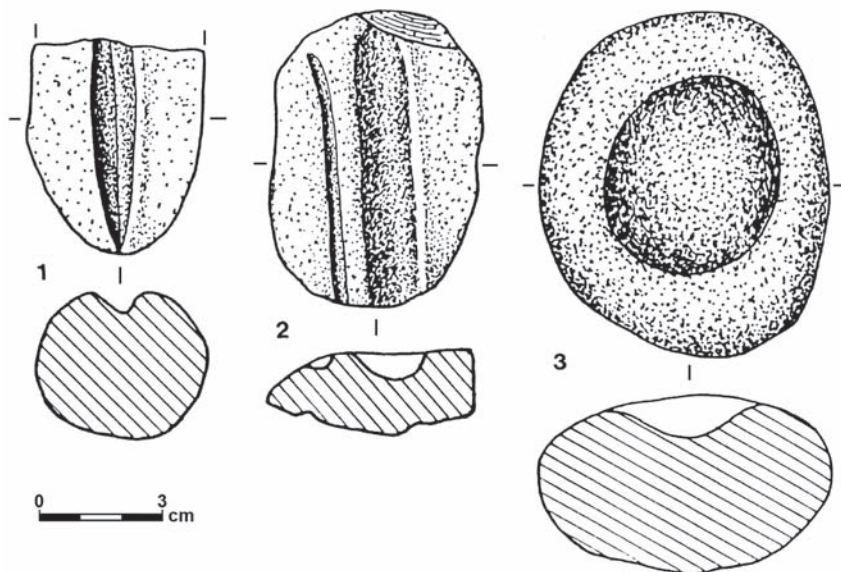
576. Jelamet el-Ahmar (B): 1-2. Axe; 3. Arrowhead type Byblos (fragment); 4. Arrowhead type Amuq; 5. Sickle blade with bifacial denticulation; 6. Cananean sickle blade.

cross-section are definitely attributed to late Pre Pottery Neolithic entities and continue to the Pottery Neolithic. The sickle blades with deep bifacial denticulation also originate from this period, e.g. Sha'ar Hagolan (Stekelis 1966); Nitzanim (Yeivin and Olami 1979); Giv'at Haparsa (Olami et al. 1977); Munhata 2b (Gopher 1989a). The fact that burins, which are scarce in later periods, are documented at Sha'ar Hagolan as well, seems to confirm that the main part of the assemblage has to be attributed to this period. The fact that 30% of the pottery also originates from this period supports this assumption.

The stone artifacts provide further evidence. The female sex organ like basalt pebble is present

at Sha'ar Hagolan (Stekelis 1966) and small cup-marks on limestone pebbles start early in the Neolithic but continue also to the later stages (Stekelis 1966; Noy 1981, 8-9).

At site 178 – Jelamet el-Ahmar (A), located about 1 km to the southeast a similar Neolithic assemblage was found. The main differences between the sites are that at site 178 the majority of the pottery originates from the Early Bronze Age, only 5% of the shards were Neolithic and no typical sickle blades were present. At Jelamet el-Ahmar (B) the presence of both Neolithic flint and pottery assemblages is considerable. A discussion about a possible connection between the two sites is beyond the scope of this study.



577. Jelamet el-Ahmar (B) – stone artifacts: 1. Basalt pebble (similar to a female sex organ); 2. Sandstone shaft sharpener; 3. Small cup mark on a limestone pebble.

Site no. 182

Mughur 'Id

Isr. Grid Ref. 1861/1810

The flint assemblage:

Early Bronze Age 1 – highly reasonable identification.

Artifacts without retouch and waste:

Primary flake 1.

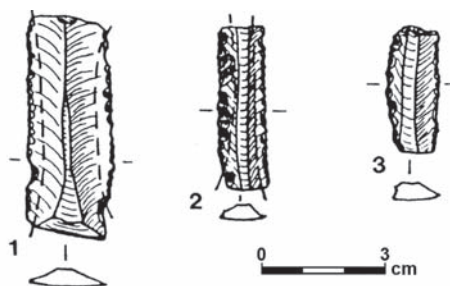
Tools and retouched artifacts:

Cananean sickle blade 2; Backed Cananean blade 1.

Total of flint artifacts – 4.

Discussion:

The small flint assemblage is in full accordance with the pottery finding attributed to Early Bronze Age I. Attention should be paid to the fact that the Cananean elements are quite narrow (see also the general remarks on flint findings above).



578. Mughur 'Id: 1-2. Cananean sickle blade; 3. Backed Cananean blade.



Site no. 183

Tel Za'anuni

Isr. Grid Ref. 1866/1802

The flint assemblage:

Early Bronze Age – highly reasonable identification.

Artifacts without retouch and waste:

CTE 1; Flake 3. Total – 4.

Tools and retouched artifacts:

Cananean sickle blade 2; Fan scraper (fragment) 1; Double notch 2. Total – 5.

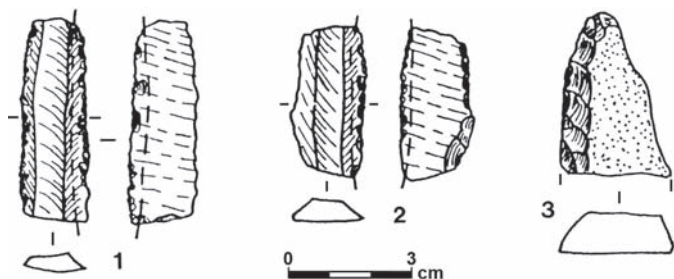
Total of flint artifacts – 9.

Discussion:

The Cananean sickle blades and the fan scraper on tabular flint with

semi-abrupt retouch are prominent tools of the Early Bronze Age (Schick 1978; Rosen 1983, 1989).

The dating of the flint findings is in full accordance with that of the pottery assemblage.



579. Tel Za'anuni: 1-2. Cananean sickle blade; 3. Fan scraper (fragment).



Site no. 185

Jelamet el-Ahmar (C)

Isr. Grid Ref. 1898/1809

The flint assemblage:

Epi-Palaeolithic Period, the Geometric Kebaran phase – highly reasonable identification.

Artifacts without retouch and waste:

Bladelet core 1; Primary flake 3; Flake 18; Blade 2; Blade (fragment) 2; Bladelet 6; Bladelet (fragment) 12; Burin spall 1; Chunk 25; Chip 43. Total – 113.

Tools and retouched artifacts:

Scrapers: End scraper on flake 1; Carinated end scraper on primary flake 2; Carinated end scraper on flake 2; Nucleiform (core) scraper 1; End scraper on snapped bladelet 2; Thumbnail scraper 1. Total

scrapers – 9.

Burins: Angled, dihedral 1; On primary flake 1. Total burins – 2.

Microlithes: Obliquely truncated bladelet 4; Obliquely truncated bladelet (fragment) 5;

Retouched bladelet (fragment) 6; Backed bladelet (fragment) 2; Arch-backed bladelet 2. Total microlithes – 18.

Geometric Microlithes: Rectangle 3; Trapeze 4; Rectangle/Trapeze (fragment) 9; Total geometric microlithes – 16.

Notch 16; Denticulate 5; Borer 4; Awl 3; Truncation 8; Retouched flake 17; Retouched blade 1; Varia 2. Total tools and retouched arti-

facts – 100.

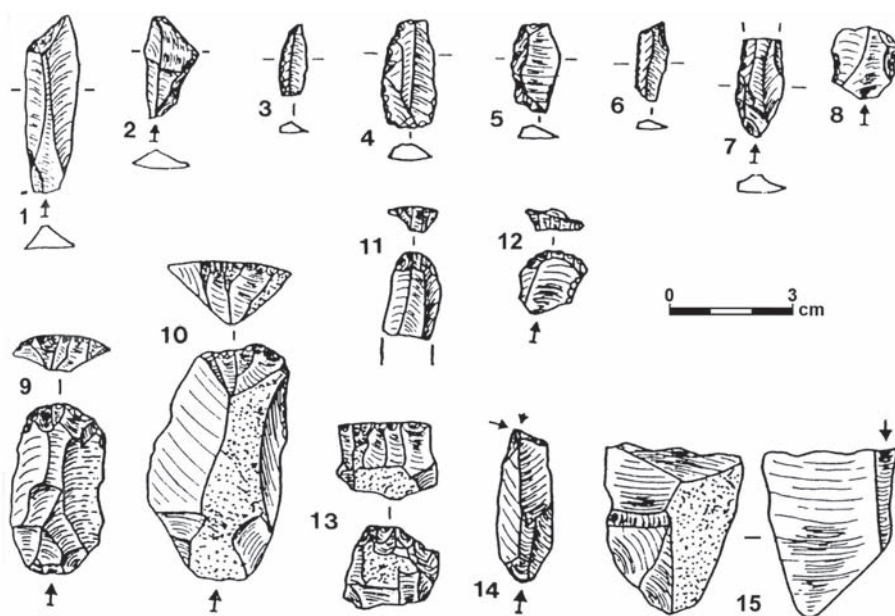
Total of flint artifacts – 213.

Discussion:

The location of the site on a hill above a stream is similar to many other Epi-Palaeolithic sites, some of them in the surveyed area: 'Iraq el-Hamra in the close vicinity and Khirbet el-Meiyiteh and Khirbet Malih (C) in Wadi Malih. Wadi Far'ah, to the north of the site, is a perennial stream until present times and to the south, close to the site according to the topography, there

were apparently some other springs flowing into Wadi Far'ah.

The flint assemblage – containing various types of scrapers, burins, microlithes and geometric microlithes – point to the Epi-Palaeolithic Period generally. The microlithes of the rectangle/trapeze group indicate the Geometric Kebaran phase. According to the quantity and the dispersal of the artifacts, it was a seasonal or temporary campsite (see also site 94 Khirbet Malih (C) for a detailed discussion).



580. Jelamet el-Ahmar (C): 1-2. Obliquely truncated artifact; 3. Arch-backed bladelet; 4. Rectangle; 5-6. Trapeze; 7. Backed bladelet (fragment); 8. Double notch; 9. Double end scraper; 10. Carinated end scraper; 11. End scraper on snapped blade; 12. Thumbnail scraper; 13. Nucleiform (core) scraper; 14. Angled burin; 15. Flat faced burin.

Site no. 189

'Iraq el-Hamra

Isr. Grid Ref. 1902/1800

Area A – The top of the hill and the southwestern slope

The flint assemblage:

Epi-Palaeolithic Period, the Natufian culture – highly reasonable identification.

Artifacts without retouch and waste:

Cores: Flake core 2; Blade core 2; Bladelet/flake core 2; Bladelet core 7. Total cores – 13.

CTE 20; Primary flake 48; Flake 134; Blade 5; Bladelet 19; Bladelet (fragment) 55; Burin spall 6; Chunk 75; Chip 63. Total of artifacts without retouch and waste – 438.

Microburin 9.

Tools and retouched artifacts:

Scrapers: End scraper on flake 9; Carinated end scraper 4; End scraper on blade 2; End scraper on snapped blade 6; *De'ete* scraper on blade 1; *De'ete* scraper on flake 3; End scraper on bladelet 4; Nucleiform (core) scraper 3; Thumbnail scraper 11; Double end scraper on flake 2; Nosed end scraper on flake 2; Narrow carinated end scraper 1; Micro-end scraper 2; Transversal scraper 1. Total scrapers – 51.

Burins: Dihedral 1; Angled dihedral 4; Double angled dihedral 2; Flat faced 1; On natural pan 3; On

straight truncation 1; On concave truncation 1; Transversal 5; Burin-scraper 1. Total burins – 19.

Microlithes: Retouched bladelet 4; Retouched bladelet (fragment) 6; Backed bladelet 2; Backed bladelet (fragment) 5; Obliquely truncated bladelet 5; Obliquely truncated bladelet (fragment) 4; Arch-backed bladelet 2; Ramon point 2; Truncated bladelet 1; Truncated bladelet (fragment) 2; Bi-truncated bladelet 1. Total microlithes – 34.

Geometric Microlithes: Proto rectangle 1; Rectangle 2; Rectangle/Trapeze (fragment) 1; Asymmetric trapeze 1; Lunate 2; Hiluwan lunate 2. Total geometric microlithes – 9.

Notch 72; Denticulate 5; Awl 10; Retouched flake 75; Backed or retouched blade 20; Truncation 6; Varia 6. Total of tools and retouched artifacts – 310.

Area A: Total of flint artifacts – 757.

Area B – The western slope

The flint assemblage:

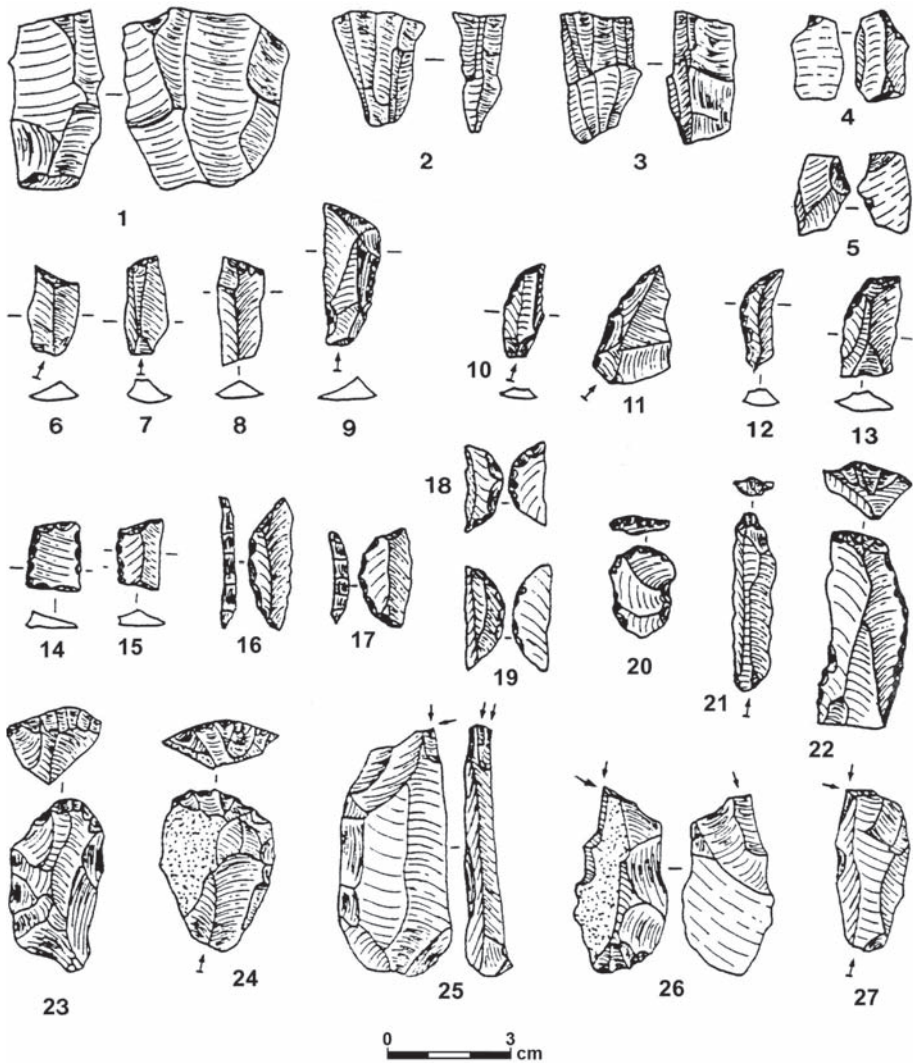
Chalcolithic Period and Early Bronze Age – highly reasonable identification.

Artifacts without retouch and waste:

Flake core 1; Primary flake 3; Flake 4; Blade 1; Blade (fragment) 3. Total – 12.

Tools and retouched artifacts:

Sickle blades: Backed, bi-truncated with fine retouch on the cutting edge 1; Backed and bi-truncated 2; Naturally backed and truncated 1; Cananean (fragment) 4. Total



581. 'Iraq el-Hamra – Area A: 1. Core; 2-3. Bladelet core; 4-5. Microburin; 6-9. Obliquely truncated piece; 10-11, 13. Arch backed piece; 12. Ramon point; 14. Rectangle; 15. Trapeze; 16-17. Lunate with abrupt backing; 18-19. Heluwan lunate; 20. Thumbnail scraper; 21. Micro-end scraper; 22-23. Carinated end scraper; 24. End scraper; 25-27. Angled burin.

sickle blades – 8.

Adze 1; End scraper on flake 1; Nucleiform (core) scraper 1; Side scraper 1; Bilateral side scraper 1; Fan scraper 1; Burin on flake 1; Notch on flake 2; Denticulate 4; Borer 4; Retouched flake 2; Backed knife 4; Backed knife (fragment) 2; Retouched blade (fragment) 2. Total of tools and retouched artifacts – 36.

Area B: Total of flint artifacts – 48.

Total of flint artifacts from both areas – 805.

Discussion:

At this site a relatively large quantity of flint artifacts was collected, mainly at the top of the hill, about 100 m from the round structure at the summit, and on the southwestern slope (area A). The area to the north and the eastern range are nearly bare of flint artifacts. On the western slope (area B), a smaller assemblage of different character was collected (see further on).

The material from area A is characterized by dark coloured flint in brown, gray and black shadings and the dimensions of both waste materials and tools are small. Waste materials include primary flakes, abundant flakes, bladelets and microburins as well. The tool group includes various types of scrapers, burins, microlithes, lunates and geometric microlithes. This composition is in accordance with Natufian assemblages. The lunate

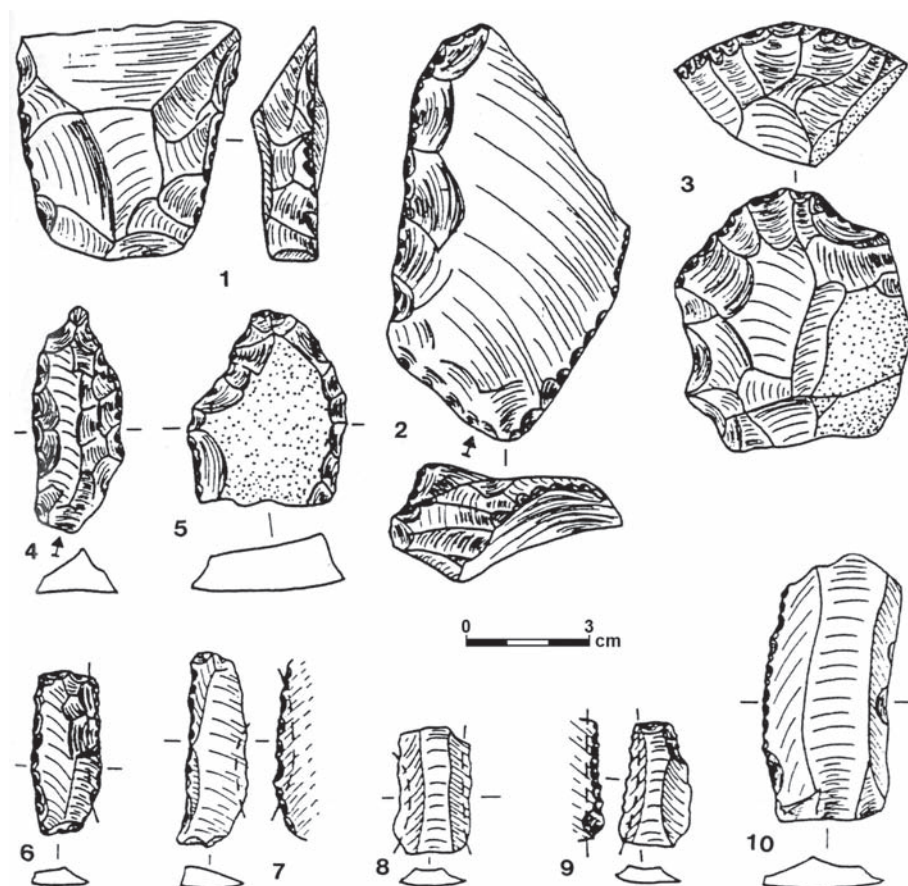
and microburin were prominent types during this period. It is quite difficult to compare the relative figures of artifacts to the material from excavated and wet or dry sieved sites (Bar-Yosef and Goren 1973, 56) because assemblages from these sites contain large numbers of small pieces and microlithes.

The fact that the assemblage contains all the components of an industry – exhausted cores, primary flakes, CTE, flakes, blades, industrial waste and various tool groups – indicates rather intensive activities at the site. However, further research is needed in order to find out whether it was a base camp or a site for repeated or seasonal camping. The location of the site, close to a perennial stream (Wadi Far'ah) is in accordance with many other sites of the period located near water sources and streams, e.g., Nahal Oren (Steckelis and Yisraely 1963; Noy et al. 1973), Einan (Perrot 1960), Poleg 18M (Burian and Friedman 1965; Bar-Yosef 1970) and Khirbet el-Meiyyiteh (site no. 90 in this volume). There is some difficulty in locating the site within the chronological sequence of the Natufian complex according to surface finds. Our tendency is to attribute the site to the earlier Natufian because of the relative width of the lunates and the presence of Heluwan retouch on two of them. This type of lunates is similar to Einan IVa (Valla 1984), el-Wad B2 (Garrod

and Bate 1937, 30-35) and different from later lunates – small, narrow and backed – similar to these from Einan C1 (Valla 1984), Rosh Horesha (Marks and Larson 1977) and Rosh Zin (Henry 1976).

The material recovered from the western slope, area B, is completely different from the material recovered from area A. Most artifacts are larger, from light coloured or calcarous flint. Some of the artifacts are

covered by a thick, white patina as visible on broken pieces. The modest number of artifacts and scarcity of cores and primary elements apparently indicate that at this location no regular flaking process was carried out. The adze, the backed sickle blades with double truncations and fine retouch of the cutting edge are prominent tools of the Chalcolithic. Adzes with a cutting edge prepared by a single blow are



582. 'Iraq el-Hamrah – Area B: 1. Adze; 2. Side scraper; 3. Carinated end scraper; 4. Borer; 5. Fan scraper; 6-7. Backed and bi-truncated sickle blade; 8-9. Cananean sickle blade; 10. Cananean blade.

not very common in the Chalcolithic, but they are present at Beer Sheva site Bir Abu Matar (Rosen, personal information). *Type fossils* for the Early Bronze Age are Canaan sickle blades; probably the fan scraper with cortex and semi-abrupt retouch should also be attributed to this period.

The artifacts recovered from the western slope testify to a human

presence of undefined character at this location during the Chalcolithic and Early Bronze Age.

No connection of the stone structures on the summit and the southwestern slope to the Natufian could be found. The few pottery shards from the western slope apparently belong to the Chalcolithic or Early Bronze Age.



Site no. 191

Mrah el-'Enab

Isr. Grid Ref. 1928/1789

The flint assemblage:

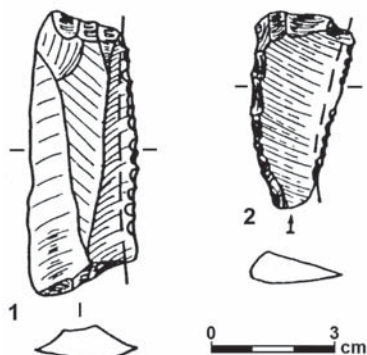
Iron Age (part of the artifacts) – Reasonable identification.

The artifacts:

Blade (fragment) 1; Geometric sickle segment 2; Ventrally retouched bladelet 1; Denticulate on flake 1. Total – 5.

Discussion:

The few artifacts do not supply sound evidence for the period. The large geometric sickle segments were in use from Middle Bronze Age II till the end of the Iron Age (Rosen 1982, 1986) but according to the pottery finds they should be attributed to the Iron Age.



583. Mrah el-'Enab: 1-2. Large geometric sickle segment.

Site no. 197

Bab ed-Dayyq

Isr. Grid Ref. 1936/1759

The flint assemblage:

Undefined.

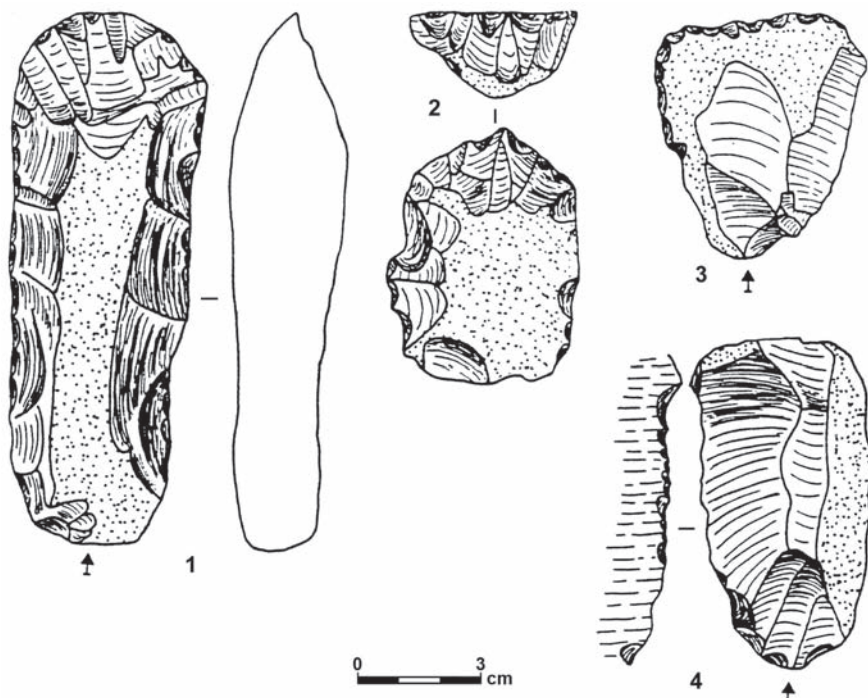
Tools and retouched artifacts:

Adze? End scraper? 1; Denticulated end scraper on primary flake 1; End scraper on flake 1; Transversal scraper on primary flake 1; Side scraper on primary flake 1; Notch 2; Double notch on fragment of sickle segment 1; Naturally backed knife with alternate retouch 1.

Total of flint artifacts – 9.

Discussion:

Most flint artifacts were from light coloured material characteristic of the later periods in this area. The majority was prepared from primary elements and generally the *ad hoc* character is obvious. The tool with the convex ventral face was produced from a large primary blade with a large portion of cortex on the dorsal face and it could be defined either as an adze or as an end scraper. The fragment of the



584. Bab ed-Dayyq: 1. Adze? End scraper? 2. End scraper; 3. Transversal scraper; 4. Naturally backed knife.

sickle segment with two notches could not be attributed to a definite period. About 20% of the pottery finds are from the Early Bronze Age but the absence of *type fossils* does

not allow a reasonable definition. To some extent it seems reasonable to attribute the assemblage to the Chalcolithic or the Early Bronze Age.



Site no. 217

Khallet el-Khanakhneh (A)

Isr. Grid Ref. 1959/1769

see Ras el-Kharube sites – Kh. III below.



Site no. 219

Ras Umm Khubezah

Isr. Grid Ref. 1977/1773

The flint assemblage:

Epi-Palaeolithic Period, the Natufian culture – possible identification.

The Neolithic Period – reasonable identification.

The Chalcolithic Period – reasonable identification.

Artifacts without retouch and waste:

Bladelet core 2; CTE 1; Primary flake 2; Flake 4; Blade 6; Bladelet 2. Total – 17.

Tools and retouched artifacts:

Backed and truncated sickle blade with retouched cutting edge 1;

Backed sickle blade with worn cutting edge 1; Reaping knife 1; Reaping knife with burin blow (fragment) 1; Geometric sickle segment ? 1; Polished axe (fragment) 1; End scraper 4; Nucleiform (core) scraper 1; Dihedral angled burin 1; Flat faced burin 1; Notch 6; Double notch 2; Denticulate 4; Awl 7; Borer 2; Retouched flake 3; Retouched blade 5; Truncation 2. Total – 44.

Total of flint artifacts – 61.

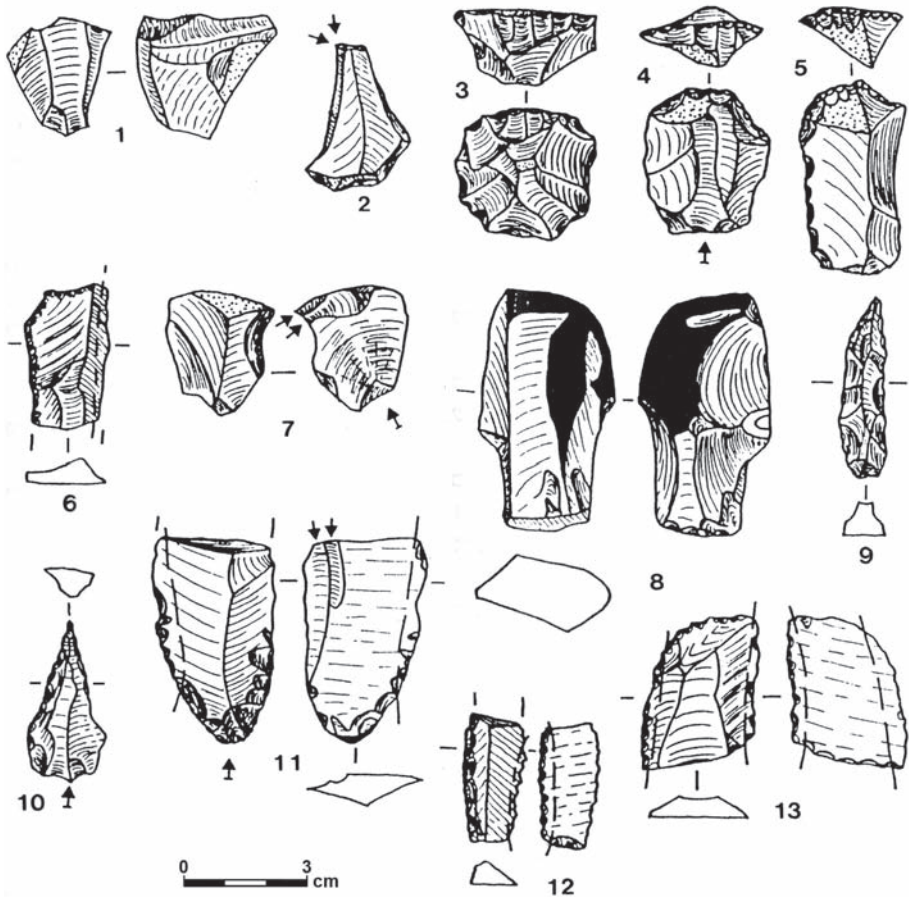
Discussion:

The assemblage represents a quite complicated picture as *type fossils* from different periods were recovered:

Without excavations and sieving it is quite difficult to find small microliths, but the bladelet cores, the nucleiform scraper on an exhausted bladelet core, end scrapers and burins point to an Epi-Palaeolithic entity. The backed sickle blade with a damaged cutting edge shows similarity to sickle blades recovered at many Natufian sites, e.g. el-Wad

B2, Kebara B, Einan B1 (after Valla 1984, 248-251, 302-303); Salibiya 1 (Crabtree and Campana 1990, 113) and could serve as a marker of the Natufian culture.

The reaping knives with re-touched cutting edge and proximal retouched tang point to the Neolithic Period – probably Pre Pottery B (Noy et al. 1983, 86-87; Gopher



585. Ras Umm Khubezah: 1. Bladelet core; 2. Angled burin; 3. Nucleiform (core) scraper; 4. End scraper; 5. Carinated end scraper; 6. Backed sickle blade (Natufian?); 7. Flat faced burin on notched flake; 8. Polished axe (fragment); 9-10. Borer; 11. Reaping knife (fragment) with burin blow; 12. Backed and truncated sickle blade; 13. Geometric sickle segment ?.

1989a, 50-64).

The Chalcolithic Period is represented by the polished axe, the backed and truncated sickle blade with a retouched cutting edge and probably also the borers (Yeivin 1958; Hennessey 1969; Lee 1973; Levi and Rosen 1987). At this stage there is no explanation for the absence of Chalcolithic pottery at the site.

The large geometric sickle segment could be attributed to later periods – from Middle Bronze Age II until the Iron Age (Rosen 1982, 142; 1983, 127). About 30% of the

pottery, at the site, was attributed to each of these periods; therefore it is impossible to attribute the sickle definitely to any of them. The bilateral sickle sheen on a geometric sickle segment is exceptional and possibly the tool was prepared from the distal end of a Cananean sickle blade.

Similarly to other sites in the region, the artifacts attributed to the later periods are of light coloured flint, while these attributed to the Natufian are smaller in size and of dark, gray shaded flint.



Site no. 225

el-‘Alam (A)

Isr. Grid Ref. 1951/1755

The flint assemblage:

Early Bronze Age – Reasonable identification.

Artifacts without retouch and waste:

Primary flake 1; Blade 3; Blade (fragment) 3. Total – 7.

Tools and retouched artifacts:

Backed sickle blade with fine retouch on cutting edge 1; End scraper 3; Double carinated end scraper 1; Side scraper 1; Notch 1; Notch on backed blade 1; Double notch on retouched flake 1; Double notch on retouched blade 2; Awl 1; Re-

touched blade 1. Total – 13.

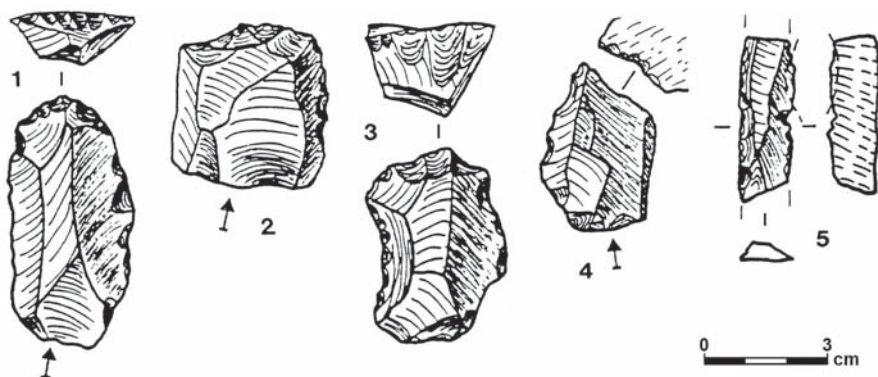
Total of flint artifacts – 20.

Discussion:

Backed sickle blades with fine retouch on the cutting edge are usually attributed to the Chalcolithic (Burian and Friedman 1979; Rosen 1983, 127, 132) but no definite conclusions should be drawn from a single artifact. The absence of bifacial tools such as axes and adzes, present in most assemblages from this period, supports the assumption that the assemblage is not Chalcolithic. At Bab edh-Dhra, about 55 km to the south, most sickle

blades from the Early Bronze Age were backed (McConaughy 1979). In Gamla too, backed sickle blades from this period were recovered (Olami 1989). The composition of the assemblage (scrapers, notches, mainly *ad hoc* tools) comply with

Early Bronze Age findings, even though *type fossils* such as Cananean blades and fan scrapers were missing. Half of the pottery finds were attributed to this age and it seems reasonable that the flint should be attributed similarly.



586. el-'Alam (A): 1-2. End scraper; 3. Double carinated scraper; 4. Awl; 5. Backed sickle blade.



Site no. 230

'Ain el-Hilu (B)

Isr. Grid Ref. 1981/1922

The flint assemblage:

Chalcolithic Period (main part of the assemblage) – highly reasonable identification.

Undefined (few artifacts).

Artifacts without retouch and waste:

Pyramidal bladelet core 1; Flake 1; Blade 2; Blade (fragment) 1. Total – 5.

Tools and retouched artifacts:

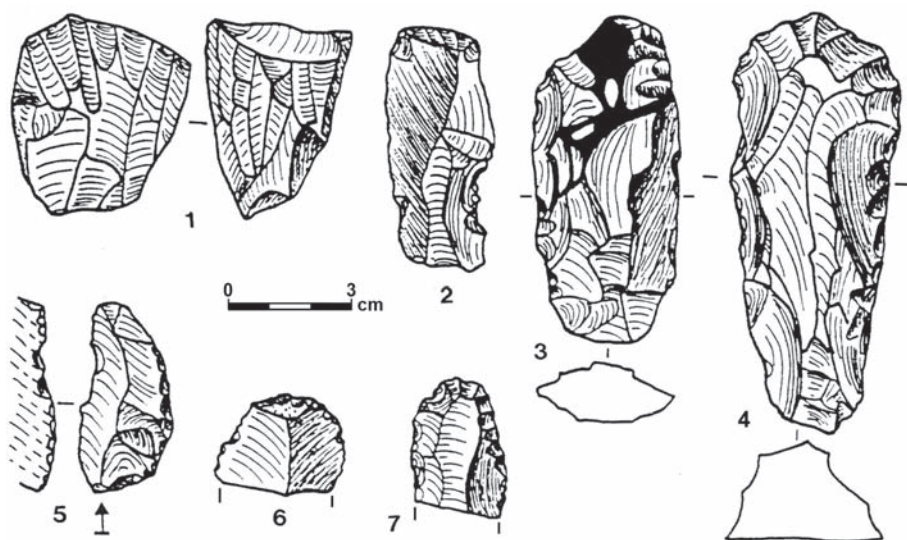
Polished adze 1; Adze (atypical on flake) 1; Carinated end scraper on blade 1; End scraper on primary flake 1; End scraper on flake 2; End scraper on snapped blade 3; Notch on flake 6; Notch on truncated blade 1; Awl 1; Borer 1; Retouched flake 5; Retouched blade 1; Truncated blade/bladelet 3; Backed knife with ventral retouch on the cutting edge 1. Total – 28.

Total of flint artifacts – 33.

Discussion:

According to the composition and technology a large part of the assemblage fits the Chalcolithic. The polished adze, the carinated end scraper on a blade with trapezoid cross-section and bilateral abrupt retouch, and the backed knife serve as prominent tools representing the period (Yeivin 1958; Hennessey 1969; Lee 1973; Roshwalb 1981;

Levi and Rosen 1987). The Chalcolithic component of the flint finds corresponds to the pottery finds from this period. Other artifacts – the thin end scrapers on snapped blades, the truncated bladelets and probably the bladelet core – could point to the Epi-Palaeolithic, but in light of the missing *type fossils* this part of the assemblage will be considered as undefined.



587. 'Ain el-Hilu (B): 1. Pyramidal bladelet core; 2. Notch on truncated blade; 3. Polished adze; 4. Carinated end scraper on blade; 5. Backed knife; 6-7. End scraper on snapped blade.

Site no. 231

el-Beyaz

Isr. Grid Ref. 2003/1926

The flint assemblage:

Undefined.

Artifacts without retouch and waste:

CTE 1; Primary flake 10; Flake with partial cortex cover 13; Flake 20; Blade 5; Bladelet 1; Burin spall 1; Chunk 7. Total – 58.

Tools and retouched artifacts:

Burin 2; Partially retouched bladelet 2; Notch 3; Retouched flake 3. Total – 10.

Total of flint artifacts – 68.

Discussion:

The composition of the assemblage indicates that the location served as a workshop for initial preparation of flint cobbles. This assumption is based on the absence of cores and chips, while the number of primary flakes and flakes partially covered

with cortex is quite high. The tool group proved marginal working and retouch, and notches could be caused by sporadic blows. Accordingly, these tools are not indicative for the identification.

The artifacts are sharp and apparently not abraded. The greater part of the artifacts are small in dimensions, from brown and gray shaded flint, partly covered by a spotted, light gray patina. These traits show a similarity to the material from Epi-Paleolithic sites in the area, such as site 94 – Khirbet Malih (C), at a distance of 4.5 km. At this site the components of the initial preparation of flint cores are missing. Further research is needed in order to verify whether el-Beyaz was the source of flint for Khirbet Malih (C).



Site no. 232

Tabqet el-Hilweh

Isr. Grid Ref. 1975/1915

The flint assemblage:

Middle Bronze Age I (main part of the assemblage) – highly reasonable identification.

Artifacts without retouch and

waste:

Primary flake 3; Flake 3; Blade 3; Chunk 2. Total – 11.

Tools and retouched artifacts:

Cananean sickle blade 4; End scrap-

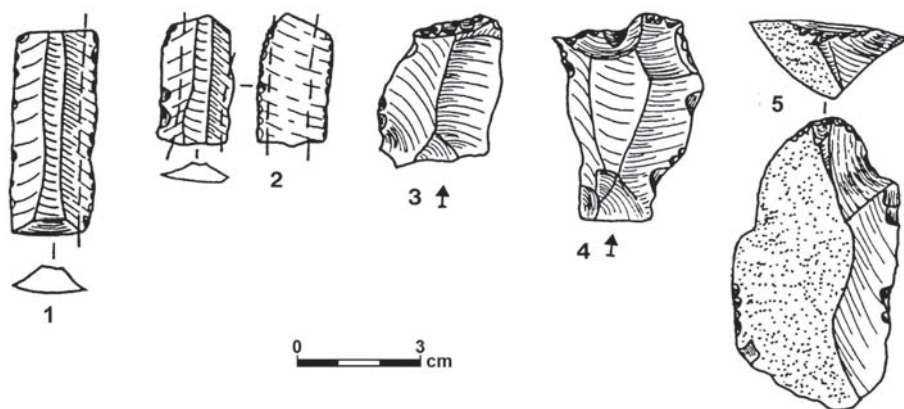
er on flake 1; Shouldered scraper on primary flake 1; Notch 7; Awl 7; Retouched flake 4; Retouched blade 4; Truncation 3. Total – 31. Total of flint artifacts – 42.

Discussion:

Similarly to other sites from the Middle Bronze Age I in the area, the presence of Cananean sickles is prominent. This period closes the chapter of the Cananean technology (Crowfoot-Payne 1983, 723; Rosen 1983, 127; 1989, 208). Here, similarly to site 66 – Wadi el-Hammeh and site 90 – Khirbet el-Meiyiteh, the Cananean sickles

are of the wider variety, mostly not truncated and breaks could also be post depositional. Following the absence of *type fossils* that might indicate other periods it can be assumed that the artifacts originate from Middle Bronze Age I. The retouched blades, borers, awls and scrapers show some similarity to these from Har Yerukham (Gilead 1973, 133), a site attributed to the same period.

Half of the pottery finds were attributed to Middle Bronze Age I and accordingly the flint finds should be attributed similarly.



588. Tabqet el-Hilweh: 1-2. Cananean sickle blade; 3. Truncation; 4. Awl; 5. Carinated, shouldered scraper on primary flake.

Site no. 234

Re'us et-Tabaq

Isr. Grid Ref. 1972/1906

The flint assemblage:

Middle Bronze Age 1 – Reasonable identification.

Artifacts without retouch and waste:

Flake 2; Blade 5. Total – 7.

Tools and retouched artifacts:

Cananean sickle blade 1; End scraper 1; Side scraper 1; Notch 1; Double truncation 1. Total – 5.

Total of flint artifacts – 12.

Discussion:

Despite the small number of artifacts, according to the Cananean sickle blade and the pottery finds (90% MBA I) the assemblage should be attributed to Middle Bronze Age I. For a detailed discussion and references, see site no. 232 – Tabqet el-Hilweh.



Site no. 246

Abu Sha'areh

Isr. Grid Ref. 1992/1849

The flint assemblage: Undefined.

Artifacts without retouch and waste:

Flake core 1; Flake 2; Blade 2. Total – 5.

Tools and retouched artifacts:

Raclette 1; Notch 2; Denticulate 1; Awl 2; Retouched flake 1; Re-

touched blade 1. Total – 8.

Total of flint artifacts – 13.

Discussion:

Eighty percent of the pottery finds were attributed to the Iron Age; however, as no *type fossils* were recovered the assemblage was classified as undefined.

Site no. 253

Wadi Abu el-Loz

Isr. Grid Ref. 1981/1820

The flint assemblage:

Epi-Palaeolithic Period, the Natufian Culture – possible identification.

Artifacts without retouch and waste:

Discoid core 1; CTE 2; Flake 4; Blade 2; Bladelet 1; Chip 3. Total – 13.

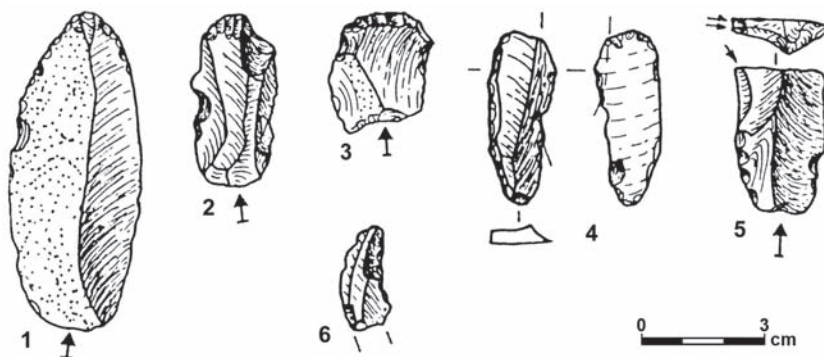
Tools and retouched artifacts:

Backed sickle blade 1; End scraper on flake 3; End scraper on primary element 1; Angled burin on flake 1; Burin on blade 1; Burin on CTE. 1; Retouched bladelet 1; Notch on ventral face 2; Notch on primary flake 1; Denticulate 1; Awl 2; Retouched flake 5; Retouched blade 5; Truncated blade 1. Total – 26.

Total of flint artifacts – 39.

Discussion:

The absence of *type fossils* in the assemblage makes an exact identification quite difficult. The end scrapers, the burins, the bladelets and the sickle blade point to the Natufian Culture like el-Wad B2, Kebara B, Einan B1 (after Valla 1984, 248-251, 302-303) and others. The assemblage was mainly produced from brown to gray flint, similar to the other Epi-Palaeolithic sites in the area. Without excavations and sieving it is quite difficult to find the typical small microlithes; therefore the identification is defined as possible. The pottery finds at the site were from historic periods and have no relation to the flint assemblage.



589. Wadi Abu el-Loz: 1-3. End scraper; 4. Backed sickle blade; 5. Angled burin; 6. Retouched bladelet.

Site no. 256

Zahret el-Meidan (B)

Isr. Grid Ref. 1992/1805

The flint assemblage:

Middle Bronze Age II – possible identification.

Artifacts without retouch and waste:

Discoid core 1; Flake 2; Blade 3. Total – 6.

Tools and retouched artifacts:

Cananean sickle blade 1; End scraper on flake 1; Notch 3; Denticulate 3. Total – 8.

Total of flint artifacts – 14.

Discussion:

The small assemblage could fit different periods, with the exception

of the Cananean sickle blade with bilateral sheen. This artifact disappeared apparently during Middle Bronze Age I (Rosen 1983, 133; 1989, 214). At Tell Nagila, a Middle Bronze Age II site, Cananean sickles still prevailed (Gilead 1973, type 3, fig. 4 no. 15) but this evidence is not sound enough to expand the period of these sickles to later periods. Ninety percent of the pottery finds were attributed to the Middle Bronze Age II and the flint assemblage should be attributed similarly. The Cananean sickle blade is an exception or its presence was incidental.



Site no. 261

Ras en-Naqb Baqar

Isr. Grid Ref. 1998/1779

see Ras el-Kharube sites – Kh. II below.



Site no. 264

Wadi Ras Umm Khubezah

Isr. Grid Ref. 1978/1764

The flint assemblage: Undefined.

Artifacts without retouch and waste:

Blade/flake core 1; Primary flake 2; Flake 6. Total – 9.

Tools and retouched artifacts:

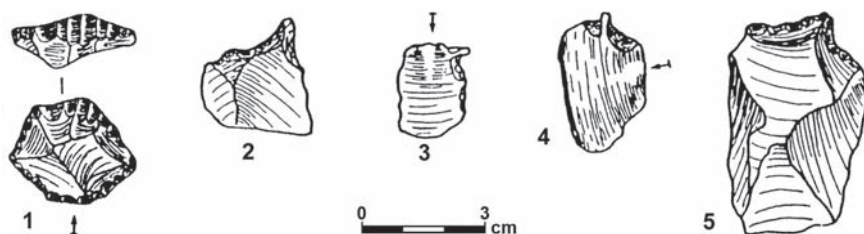
Thumbnail scraper 1; Straight side scraper 1; Notch 1; Double notch 1; Denticulate 1; Borer 1; Needle awl 4; Retouched flake 9; Truncation 2. Total – 21.

Total of flint artifacts – 30.

Discussion:

Generally the flint artifacts are not abraded and sharp. The flint quality varies; some items are from light coloured calcareous flint and others are from high quality brown,

gray or reddish flint. Missing *type fossils* and the character and size of the assemblage make it impossible to reach any chronological conclusions. Worthy of mentioning are the four needle awls similar to these from Hatula (Ronen and Lechevallier 1985) and Netiv Hagdud (Nadel 1988), and the short, thick borer prepared from a fragment, propably of a Chalcolithic axe or adze, with trapezoid cross-section.



590. Wadi Ras Umm Khubezah: 1. Thumbnail scraper; 2-4. Needle awl; 5. Borer.



Site no. 268

el-Mas'udi

Isr. Grid Ref. 1969/1733

The flint assemblage:

Wadi Rabah phase (final Pottery Neolithic) – highly reasonable identification.

Artifacts without retouch and waste:

Primary flake 4; Flake 23; Blade 3; Blade fragment 3; Bladelet 1; Chunk 4. Total – 38.

Tools and retouched artifacts:

Backed and bi-truncated thick sickle blade (type D, Gopher 1989a) 6; Naturally backed sickle blade 1; Backed and bi-truncated blade (similar to type D sickle) 3; Bi-truncated blade with bilateral retouch 1; Naturally backed, retouched blade (like sickle) 2; End scraper on flake 4; Carinated end scraper 1; Thumb-

nail scraper 1; Nosed end scraper 1; Double shouldered end scraper 1; Transversal scraper 1; *De'ete* scraper 1; Notch 4; Double notch 1; Denticulate 2; Awl 2; Borer 1; Massive borer 1; Retouched flake 6; Truncation 3; Backed and truncated knife 1; Varia 2. Total – 46.

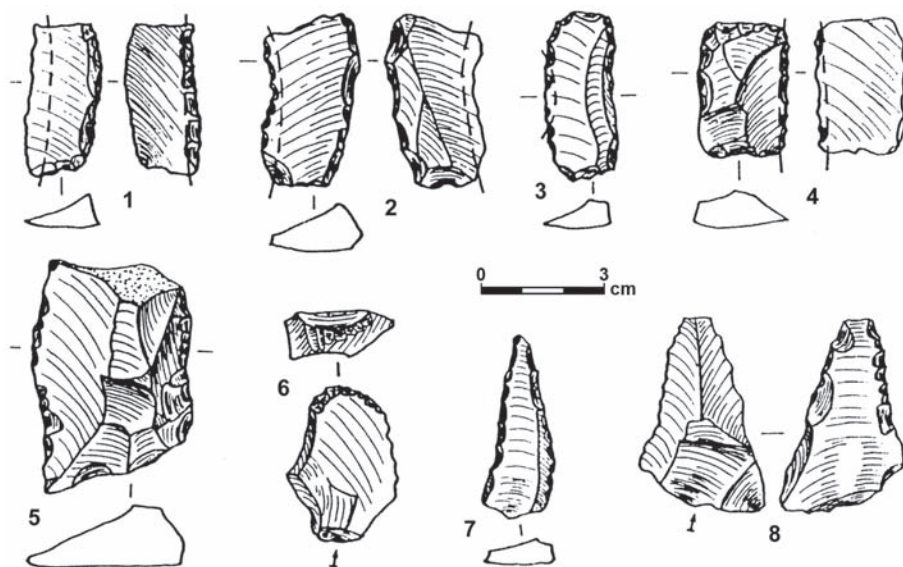
Total of flint artifacts – 84.

Discussion:

The flint artifacts were prepared from brown and gray, good quality flint. Some of them were covered by a spotted light gray patina. Few artifacts showed newer retouch without patina, perhaps indicating reuse of ancient pieces, apparently from the Middle Palaeolithic Period. The sharpness of the artifacts indicates that they were found *in situ*.

The *type fossil* for the period is the backed and bi-truncated thick sickle blade with quite coarse retouch on the cutting edge (either dorsal or ventral). These, and similar blades without sickle sheen, compose about 15% of the assemblage. These tools are characteristic or even unique to the Wadi Rabah Phase. Furthermore, the massive borer also supports this assumption (Gopher 1989a) and all the other tools fit well into the frame of the period.

Small body shards of pottery partly covered with red paint, a ribbon handle and the base of a V shaped vessel were also recovered. All these comply with the attribution to the Wadi Rabah phase.



591. el-Mas'udi: 1-4. Backed and truncated thick sickle blade; 5. Backed knife; 6. End scraper on flake; 7. Borer; 8. Massive borer.

Site no. 269

el-Makhruq

Isr. Grid Ref. 1983/1707

The flint assemblage:

Pottery Neolithic Period – highly reasonable identification.

Wadi Rabah phase – highly reasonable identification.

Chalcolithic Period – highly reasonable identification.

Early Bronze Age – highly reasonable identification.

Artifacts without retouch and waste:

Core 17; CTE 1; Ridge blade 2; Primary flake 30; Flake 69; Blade 24; Blade fragment 8; Bladelet 3; Chunk 31; Chip 3; Hammerstone 2. Total – 190.

Tools and retouched artifacts:

Backed and bi-truncated sickle blade with fine retouch on cutting edge 5; Backed sickle blade (fragment) 1; Sickle blade, unretouched (fragment) 1; Bi-truncated sickle blade with bifacial pressure flaking 1; Cananean sickle blade 6; Cananean sickle blade (fragment) 3; Bifacial polished axe 1; Axe – cutting edge fragment 1; End scraper on flake 1; *D'ete* end scraper 1; Carinated end scraper 6; Nucleiform (core) scraper 4; Side scraper 4; Transversal scraper 4; *D'ete* side scraper 1; Fan scraper with carvings 1; Fan scraper with carvings (fragment) 1; Burin 3; Retouched bladelet 2; Notch 34;

Double notch 3; Denticulate 10; Borer 6; Massive borer 2; Borer on fragment of bifacial tool with trapezoid cross-section and bilateral polish 1; Awl 13; Retouched flake 22; Retouched blade 5; Retouched blade (fragment) 6; Backed blade 5; Cananean blade 5; Truncation 4; Backed knife 1; Varia 10. Total – 176.

Total of flint artifacts – 366.

Stone tools:

Bifacial polished basalt axe 1; Perforated discoid tool, from hard limestone (fragment) 1; Limestone shaft sharpener 1; Basalt vessel (rim fragment) 1.

Discussion:

The flint assemblage provides quite a complicated picture. Artifacts from four different periods were recovered; the earliest is the Pottery Neolithic Period. A possible *type fossil* of this period is the bifacial pressure flaked (Abu Gosh flaking) sickle blade with lentoid cross-section and sickle sheen on both faces. This kind of retouch was first recognized on artifacts from the Late Pre Pottery Neolithic and continued to the Pottery Neolithic. Similar blades were recovered at Giv'at Haparsah (Olami et al. 1977), Nit-zanim (Yeivin and Olami 1979), and probably also in Jericho (Pot-

tery Neolithic, type B – Crowfoot-Payne 1983). Apparently the burins should also to be attributed to this period, similarly to those from Sha'ar Hagolan (Steckelis 1966), Giv'at Haparsah and Nitzanim. During the Wadi Rabah phase the share of burins in assemblages diminishes (except Nahal Zehora I – Gopher and Orrelle 1990) and from the Chalcolithic Period on they disappear nearly completely. Another typical tool is the backed knife which made its reappearance during the Pottery Neolithic (Gopher 1989a) and continues to the Wadi Rabah Phase and the Chalcolithic. In conclusion, most of the mentioned artifacts should be attributed to the Pottery Neolithic. This conclusion is in accordance with the pottery finds from the excavation at area C (Gimel in Hebrew – Eisenberg 1992). Polished basalt axes are reported from earlier Neolithic phases and are not known in the Pottery Neolithic, but no conclusions should be made from a single artifact.

The massive borers (Kaplan 1969; Gopher 1989a; 1989b; Gopher and Orrelle 1990), the borer on a fragment of a bifacial tool with trapezoid cross-section and a part of the backed and truncated sickle blades fit well to the Wadi Rabah tool kit. The identification of this phase is supported by the fact that at the site Wadi Rabah pottery shards were found (Gopher, per-

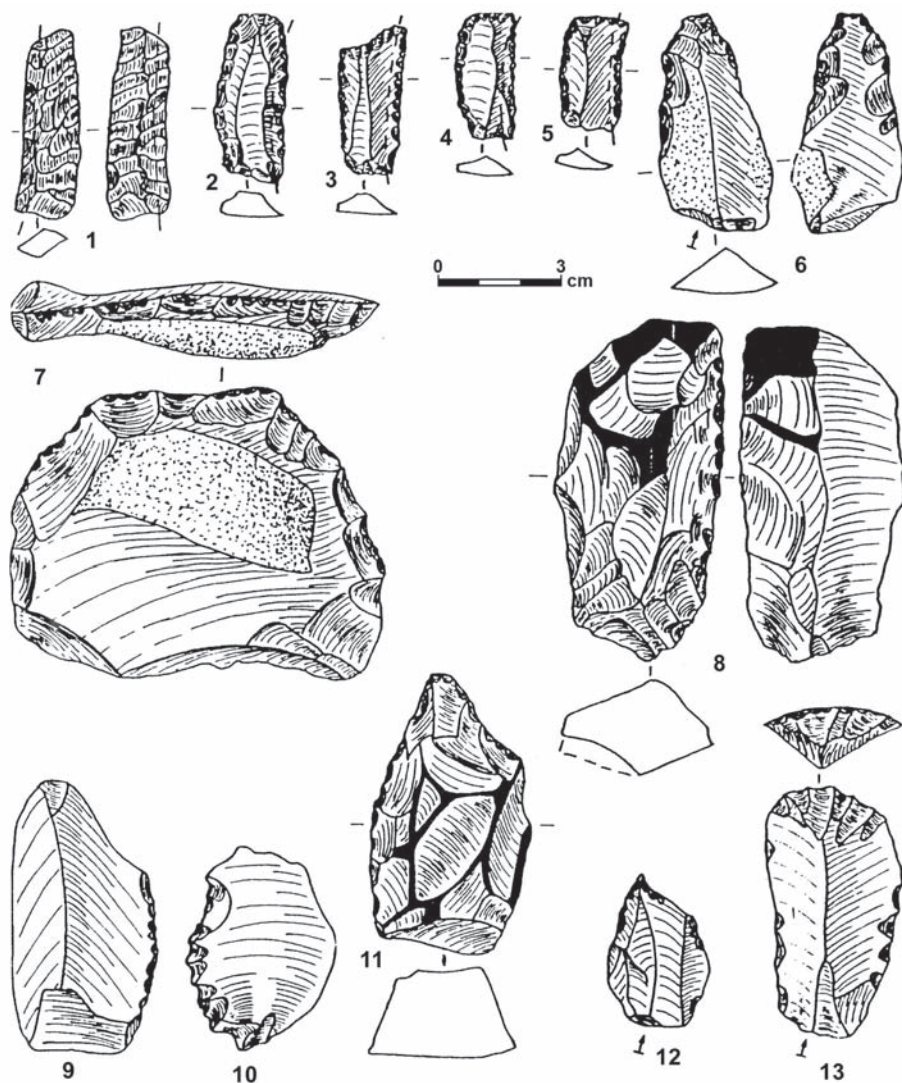
sonal information).

A definite *type fossil* for the Chalcolithic Period is the perforated discoid tool which was recovered at many sites from the south to the Golan in the north. These specific tools are totally absent in Wadi Rabah sites (Gopher, personal information; Kaplan 1969; 1977). At el-Makhruq a large fragment of such a perforated tool was recovered (about 15 cm diameter and 3.9 cm thickness in the center; see Fig. 593/1). Most of the circumference was shaped by alternate detaching of coarse flakes, and a more-or-less flattened base was prepared by bifacial abrupt retouch. The hole in the center is symmetric, rounded and was shaped carefully by gentle, point-like chipping. Morphologically the tool resembles such tools from Neveh Ur (Perrot et al. 1967) and type IV (Epstein and Noy 1988). The artifact was prepared from hard limestone similarly to such artifacts from Abu Matar (Perrot 1955, 78). Other tools which could be attributed to the Chalcolithic are the polished axe with a triangular cross-section (later a longitude blade was detached) and probably the fan scraper (Fig. 592/7) with cortex cover (without carvings on the cortex) and coarse retouch (Yeivin 1958; Levy and Rosen 1987) and the backed and bi-truncated sickle blades with fine retouch on the cutting edge (eventually some of these could also be at-

tributed to the Wadi Rabah phase).

The fourth period represented in the flint assemblage is the Early Bronze Age. *Type fossils* for this period are the fan scrapers with cortex cover and carvings on the cortex

and carefully prepared bases (Fig. 593/2-3), and the Cananean blades and Cananean sickle blades (Schick 1978; Crowfoot-Payne 1983; Rosen 1989). The fan scrapers resemble those recovered in Arad,



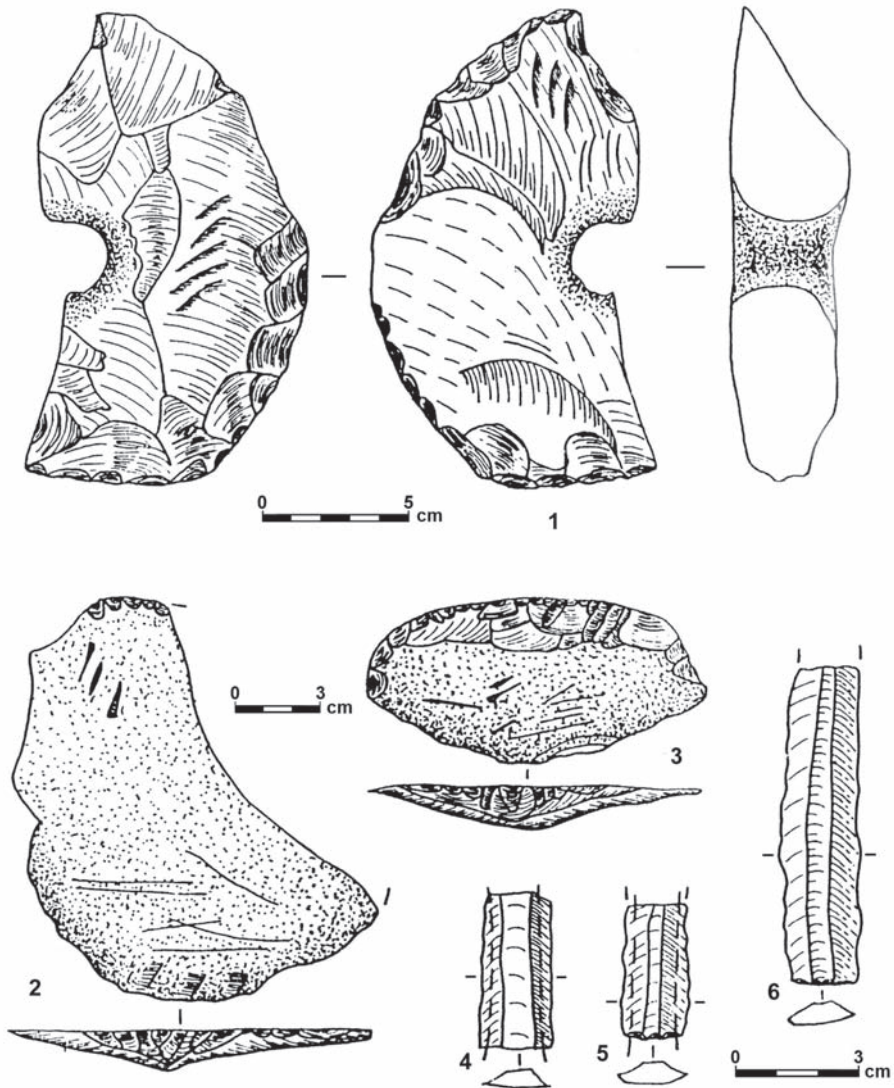
592. el-Makhruq (part 1): 1. Sickie blade (pressure flaking); 2-5. Backed and truncated sickie blade; 6. Massive borer; 7. Fan scraper; 8. Polished axe; 9. Backed knife; 10. Denticulate; 11. Borer on bifacial tool fragment; 12. Awl; 13. End scraper.

levels II-IV (Schick 1978). This resemblance can be observed by the carvings on the cortex, the preparation of the bases and the similarity of the flint.

Other tools which could originate from more than one period

were of no use for the identification of the periods.

Generally the flint assemblage is in accordance with the pottery finds from the survey and the excavations.



593. el-Makhruiq (part 2): 1. Perforated discoid tool; 2-3. Fan scraper with carvings; 4-5. Cananean sickle blade; 6. Cananean blade.

Sites at the Ras el-Kharube massif and its vicinity:

(See also the introduction and the prologue)

The flint finds recovered from Ras el-Kharube and its vicinity point to a group of sites from the Middle Palaeolithic Period. A group of single-period sites located in a limited geographical area needs a different system of analysis than the verbal description of each site applied for the other sites in this annex. In order to create a clear picture, a comparison of the frequency of the various artifact groups in the assemblages from the studied area was carried out (tab. 3). Furthermore the indices of some assemblages were compared to those from some excavated contemporaneous sites (tab. 5). The limited sample size impedes carrying out a full analysis according to the conventional system of Bordes. Accordingly, only the main groups of Bordes were reported and few indices calculated (tables 4-5 below). The value of these comparisons is limited because of the difficulty in comparing excavated sites and sporadic surface collections. Furthermore, in our case the thickness of the flint bearing horizon is unknown and altogether the picture should be considered as partial and preliminary.

On the Ras el-Kharube massif, many Middle Palaeolithic artifacts are dispersed all over the area, not only at the listed sites or find spots. Most flint concentrations in this

area are at locations different from the surveyed sites from other periods and they were numbered differently – Kh. I (Kharube I) to Kh. XI; the coordinates are according to the Israel Grid Reference and mark the center of the collected area.

Discussion and evaluations:

The studied assemblages prove similarity to each other in the composition of tool and waste material groups and flint varieties. The artifacts are mostly sharp, not abraded and heat cracking is rare. The assemblages are homogenous and no artifacts which could be attributed to other periods were recovered in this area. The composition, of both the tool and waste groups, points definitively to a flake industry.

The raw flint originates from cobbles imbedded in the limestone layers of the Zor'a formation from the Eocene Age protruding at the northwestern fringes of the Ras el-Kharube massif. Some patinated artifacts had regular retouch without patina, and originated apparently from a later phase, indicating a long term occupation of the area within the period.

The ratio of cores, waste materials and tools differs considerably from the ratio in excavated sites, that as a result of the selective collection (or in few cases even unintentional

retouch or edge damage caused by natural forces). These facts have some effect on the general index (IL) and its value is therefore limited. At all sites or find spots, cores (mostly Levallois), and a wide variety of tools corresponding generally to the type list of Bordes were recovered. A still unexplained phenomenon is the low number of flakes and points produced by the Levallois technique compared to the high number of Levallois cores. Probably here too the influence of selective collection, with preference for larger, easily visible artifacts is recognizable. At all 11 flint concentrations on Ras el-Kharube, cores, flakes, blades, industrial debris and a variety of retouched tools were recovered. A metric analysis was not carried out but it should be emphasized that no elongated Levallois points were located.

The density of artifacts varies from site to site and without systematic excavations and research it is impossible to define the character and the extent of activities at each of them.

The find spot Kh. IV, grid ref. 1958/1767, differs from the other flint concentrations. At this location, on a steep slope, flint cobbles which had been quarried from the flint lenses imbedded in the limestone matrix were found. These cobbles could serve for the preparation of cores, even though some of them were not utilized. Cores, mainly

large not exhausted Levallois cores were also present. The tools recovered proved only marginal retouch or a single burin blow, they were not completed and were apparently discarded. We assume that here was one of the sources of raw flint and a workshop for the preparation of cores and tools. A large variety of artifacts and tools indicating a wide range of activities was recovered on the ridge above the mentioned slope (find spot Kh. III, grid ref. 1959/1769).

Also from find spots Kh. VI (grid ref. 1976/1750), Kh. X (grid ref. 1975/1737) and Kh. XI (grid ref. 1968/1756) rather numerous assemblages were recovered. At these three find spots and the above mentioned find spot, Kh. IV, the largest variety of tools and unretouched artifacts was found. They all were located at elevated points overlooking the surrounding area. At the other find spots the number of artifacts and the variety of types is lower. Some of these sites were surveyed during earlier phases of the survey when the attention paid to flint artifacts was still inadequate.

At all concentrations or find spots, components characteristic for the Middle Palaeolithic Period were present. Technologically this included the Levallois technique, the preparation of the striking platforms and faceted bases and a flake based industry generally. Typologically the assemblages included vari-

Grid Reference	1980/ 1781	1998/ 1779	1959/ 1769	1958/ 1767	1965/ 1763	1976/ 1722	1956/ 1750	1972/ 1750	1972/ 1746	1975/ 1737	1968/ 1756
	Kh. I	Kh. II	Kh. III	Kh. IV	Kh. V	Kh. VI	Kh. VII	Kh. VIII	Kh. IX	Kh. X	Kh. XI
The artifacts	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
Artifacts without retouch and waste											
Levallois core	(1)	(1)	(2)	(7)	(3)	(1)	(2)	(2)	(2)	(6)	(6)
Total cores	1	1	2	11	5	5	2	2	2	7	10
Industrial waste	-	-	5	8	11	6	13	9	1	18	17
Primary flake	-	-	13	19	3	4	2	6	4	11	3
Flake	5	3	19	36	29	13	18	13	11	86	40
Blade/ Bladelet	4	1	6	8	4	1	6	1	1	12	3
Total waste	10	5	45	82	52	29	41	31	19	134	73
Levllouis pieces and tools (after Bordes)											
1 - 2	4	1	3	4	8	6	5	-	-	11	13
3	-	-	1	4	1	-	-	-	1	6	-
4	1	-	1	2	-	2	3	-	-	5	3
5	1	-	-	2	-	-	-	-	1	1	-
6 - 7	-	-	1	-	-	-	-	-	-	-	-
9 - 29	1	1	10	2	2	4	1	2	3	10	7
30 - 31	1	2	4	3	4	6	2	-	1	13	11
32 - 33	1	-	1	5	-	3	-	2	1	4	6
34 - 35	1	1	2	5	2	8	1	-	2	7	13
36 - 37	-	-	2	6	-	-	-	1	-	1	2
38	-	-	-	-	-	2	1	-	-	4	1
39	-	1	-	-	-	-	-	-	-	2	1
40	-	1	2	2	-	2	-	-	-	4	-
42	2	2	11	16	9	8	5	5	2	35	18
43	1	-	7	5	1	2	3	-	-	2	15
45 - 50	-	2	5	7	7	9	2	5	1	18	25
51	-	-	-	-	-	-	-	-	-	1	-
54	-	-	3	7	-	2	-	1	-	3	4
59 - 61	-	-	-	4	-	-	-	-	-	-	3
62	1	-	1	-	2	1	-	-	-	6	5
Total tools	14	11	54	74	36	55	23	16	12	133	127
Total artifacts	24	16	99	156	88	84	64	47	31	267	200
Levallois technique	10	2	15	13	12	9	11	3	3	33	10
Facetted platform	5	1	7	2	2	9	3	2	1	15	7

Table 3: Frequency of flint finds at Ras el-Kharube sites

Captions for table 3:

1-2	Levallois flake	38	Naturally backed knife
3	Levallois point	39	Raclette
4	Retouched Levallois point	40	Truncation
5	Pseudo Levallois point	42	Notch
6-7	Mousterian points	43	Denticulate
9-29	Side scrapers	45-50	Retouched pieces
30-31	End scrapers	51	Denticulated point (Tayak point)
32-33	Burins	54	End notch
34-35	Perforators	59-61	Chopping tools
36-37	Backed knives	62	Varia

Grid reference /Site	1959/1769 Kh. III	1976/1722 Kh. VI	1975/1737 Kh. X	1968/1756 Kh. XI
Relative quantities for the comparison of sites %				
Levallois pieces (from total of tools and unretouched) 1 - 62	27.78	16.36	24.36	7.87
Levallois pieces (from total less cores and waste)	16.30	12.33	13.64	5.78
Facetted platform (from total of tools and unretouched) 1 - 62	12.96	16.36	11.28	5.51
Relative quantities, Index after Bordes %				
IL ty Levallois typological (restricted)	2.27	5.26	5.38	3.49
IL ty Levallois typological (real)	9.26	14.55	16.54	12.60
II Mousterian (restricted)	25.00	10.53	10.75	8.14
II Mousterian (real)	20.37	7.27	7.51	5.51
III Upper Palaeolithic (restricted)	20.45	50.00	23.66	37.21
III Upper Palaeolithic (real)	16.67	34.55	16.54	25.20
Levallois (IL)	4.35	8.20	7.02	7.51

Table 4: Relative quantities of artifact groups and indices at Ras el-Kharube sites

ous types of points and side scrapers from the Mousterian (group II) and various types of end scrapers, burins, borers and knives from the Upper Palaeolithic (group III). Furthermore, notches and denticulates were also present in abundance.

For a comparison some indices for the four larger sites were calculated (tab. 4). Similarly to most contemporaneous sites, table 4 points to very low ILty (restricted) and ILty (real) indices. Mousterian index (group II) is low as well, especially at Kh. XI. At this site the percentages of Levallois pieces and

facetted bases are also very low. It is difficult to determine whether the reasons were functional or there were different modes along the axis of time.

Upper Palaeolithic index (group III) is dominant at Kh. VI, Kh. IX and Kh. X. Only at Kh. III it is somewhat lower than the Mousterian index. Also at all the other sites not included in table 4 the Upper Palaeolithic index is higher than the Mousterian index.

Because of the exceptional ratio between unretouched artifacts and waste and retouched artifacts

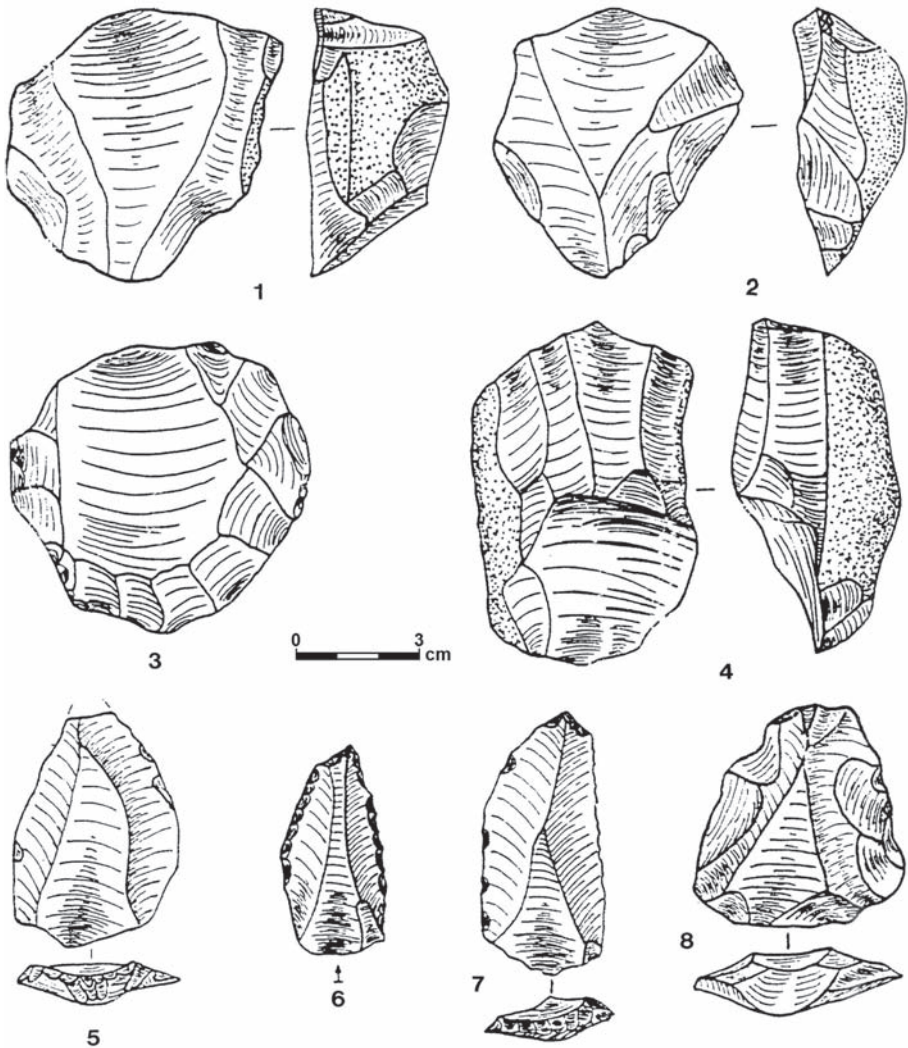
The sites	IL ty%		Mousterian Group II		Upper Palaeolithic Group III	
	Restricted	Real	Restricted	Real	Restricted	Real
Ras el-Kharubeh sites	2.27-5.38	9.26-16.54	10.75-25.00	7.27-20.37	20.45-37.21	16.54-25.20
Exceptional values			(8.14)	(5.51)	50.00	34.55
Tirat Carmel (the triangle)	1.12	16.01	34.90	23.36	20.49	13.13
Tirat Carmel (level 2-4)	2.04	7.96	26.50	23.00	22.44	19.46
Rosh Ein Mor	5.30	60.60	10.80	3.30	30.40	9.20
Nahal Aqev (D 35)	1.75	45.54	20.18	10.27	19.30	9.82
Far'ah II	0.00	39.81	5.65	3.40	4.84	2.91
Qouneitra (A)	0.73	10.17	30.55	24.54	26.84	21.70
Qouneitra (B)	0.24	7.19	36.52	33.17	11.47	10.42

Table 5: Comparison of Ras el-Kharube sites to some open-air or excavated sites. Reference for figures: Ronen 1974; Crew 1976; Munday 1977; Gilead 1980; Goren-Inbar 1990.

and tools, caused by the collecting techniques, the Levallois index (IL) is low at all find spots Kh. I – Kh. XI and should be related to very carefully. Its value is close only to that of Far'ah II (Gilead 1980, 56) – 4.48%. Table 5 demonstrates that

no other contemporaneous site is similar to Ras el-Kharube sites in all components; therefore it is preferable to compare the restricted typological indices of the tool groups.

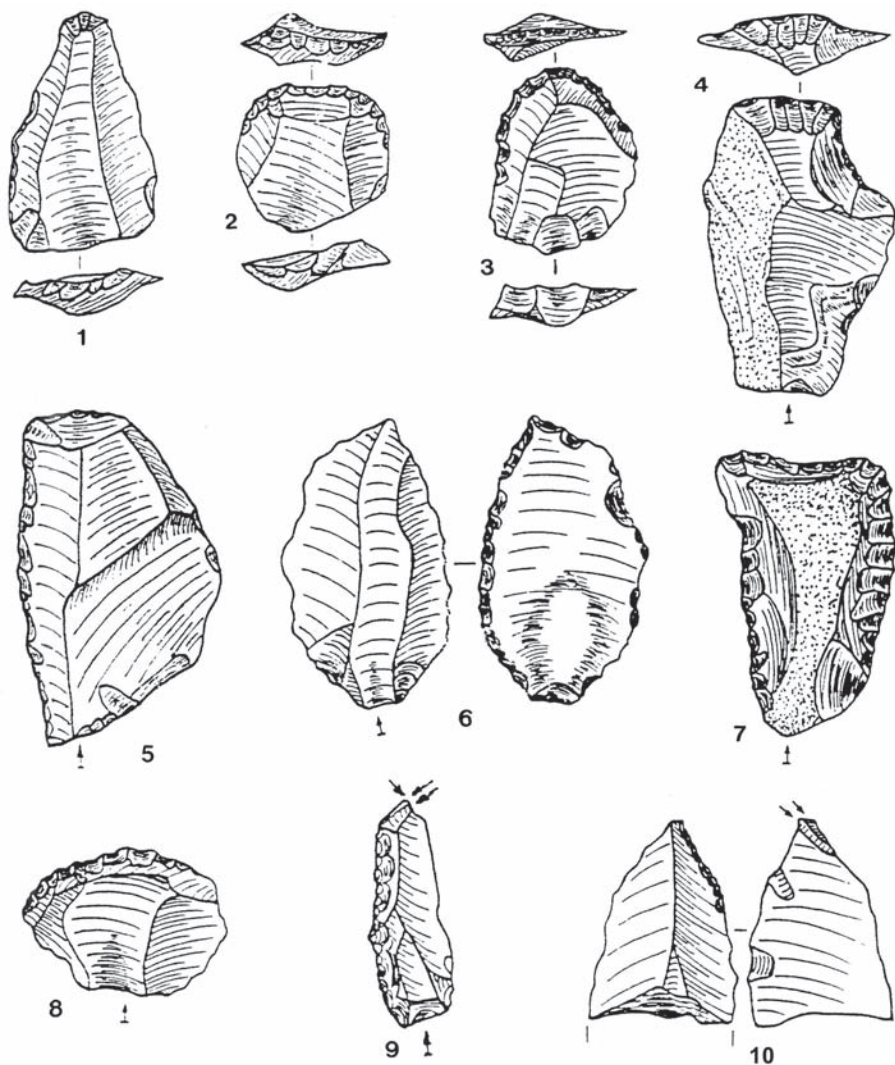
The greatest similarity of Ras el-Kharubeh restricted indices, of



594. Ras el-Kharube (part 1): 1. Levallois point core (Kh. IV); 2. Levallois point core (Kh. XI); 3. Discoid core (Kh. II); 4. Levallois blade core (Kh. III); 5. Levallois point (Kh. XI); 6. Mousterian point (Kh. III); 7. Retouched Levallois point (Kh. III) 8. Tayac point (Kh. X).

the Mousterian (group II) and the Upper Palaeolithic (group III) is to these of Rosh Ein Mor in the Negev highlands. Chronologically the excavator classified this site to a cold and humid phase of the last Ice Age

50-60 ka BP (Crew 1976, 110; see also the important remark in the prologue). In the same area the Nahal Aqev site (Munday 1977, 35-36) was similarly classified. Since these sites were published, new



595. Ras el-Kharubeh (part 2): 1-2. End scraper (Kh. X); 3. End scraper (Kh. I); 4. End scraper (Kh. III); 5. Side scraper (Kh. IV); 6. Side scraper with ventral retouch (Kh. I); 7. Bilateral side scraper (Kh. III); 8. Transversal scraper (Kh. IV); 9. Burin (Kh. II); 10. Burin (Kh. I).

methods of radiometric dating have been developed and the analyses have proved that Middle Palaeolithic sites were much earlier than first assumed. Possibly the Rosh Ein Mor and Nahal Aqev sites are also much older than assumed by the excavators.

The climatic and topographic settings of Ras el-Kharube – a mountainous, semi-desert area – are similar to those of the mountainous Negev and probably living conditions were also similar. The

similarity of the assemblages and the similar living conditions indicate, to some extent, that Ras el-Kharube sites and the mountainous Negev sites of Rosh Ein Mor and Nahal Aqev could have existed during the same time span. According to the limited sample size and partial comparisons further thorough research is essential in order to confirm or disprove this assumption. Such a research is beyond the scope of this study.

Prologue and final remarks for the English edition

1. During the periods studied, the climate differed and fluctuated over time. From about 65 ka to 15 ka BP, a period relevant for this study, Wadi Far'ah and large parts of the area were flooded by Lake Lisan (figs. A and B below). This lake is the last of several lakes which existed in the Jordan Rift Valley earlier in the Pleistocene. Following a series of fluctuations, the lake reached its maximum elevation of 164 m below present sea level about 27 ka BP (Bartov et al. 2002, 20, fig. 7 and fig. 596 in this study). It is difficult to locate, on the present surface, *in situ* prehistoric finds from periods earlier than 15-17 ka BP and lower than 164 m below sea level, because no sites could exist in the flooded area. Furthermore, over the millennia, marls (the Lisan marls) were deposited at a depth of up to

40-50 m (Bartov et al. 2002, tab. 2, fig. 6), thus earlier sites (including those which existed prior to 65 ka BP) were flooded and buried later by the lake deposits. Artifacts from these earlier sites may be found on the surface only whenever they were unearthed by natural forces, by human activity or at sites close to the shoreline, flooded only over a short period. The map (fig. 597 in this study) demonstrates clearly that Middle and Upper Palaeolithic sites, located by the surface survey, were above, and a few only close to the ancient shoreline. This refers to the sites north of Wadi Far'ah, published in this appendix, and these published in volume IV south of Wadi Far'ah. Actually, in the lower areas, most surface finds and sites were from the later periods, from the Epi-Palaeolithic and younger pe-

riods.

2. The date suggested for the Middle Palaeolithic site of Rosh Ein Mor pointed to a quite late date (Crew 1976, 76-78). In appendix 1 of Volume II it was predicted that this evaluation could possibly change after further analyses were carried out. This is exactly what happened and in 2003, a reliable Th/U analysis of ostrich eggshell dated Rosh Ein Mor somewhat earlier than 200 ka BP (Rink et al. 2003, 200).

3. The common view that in the Levantine Middle Palaeolithic no bifacial handaxes were present should be rejected because in three sites from this period, published in Volume IV, such artifacts were present (Winter 2005, sites 197, 198, 203). This fact was also confirmed by the findings from the Tabun cave

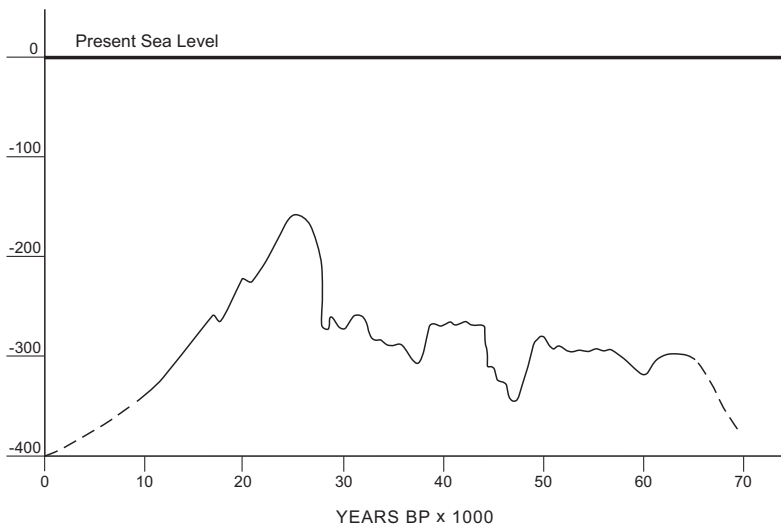
(summarized by McPherron 2003, tab. 3.1, 58).

4. The presence or disappearance of certain tools in a chronologically defined period should be treated very carefully. As mentioned above, bifacial hand axes continued to some extent in the Middle Palaeolithic Period; burins were also recovered in assemblages later than the Neolithic and Cananean sickles were found in Middle Bronze Age II sites as well.

5. The Wadi Rabah phase is today attributed definitely to the Late Pottery Neolithic.

6. Actually the Upper Palaeolithic site of Rujm ez-Zia was not published in Volume III but in Volume IV.

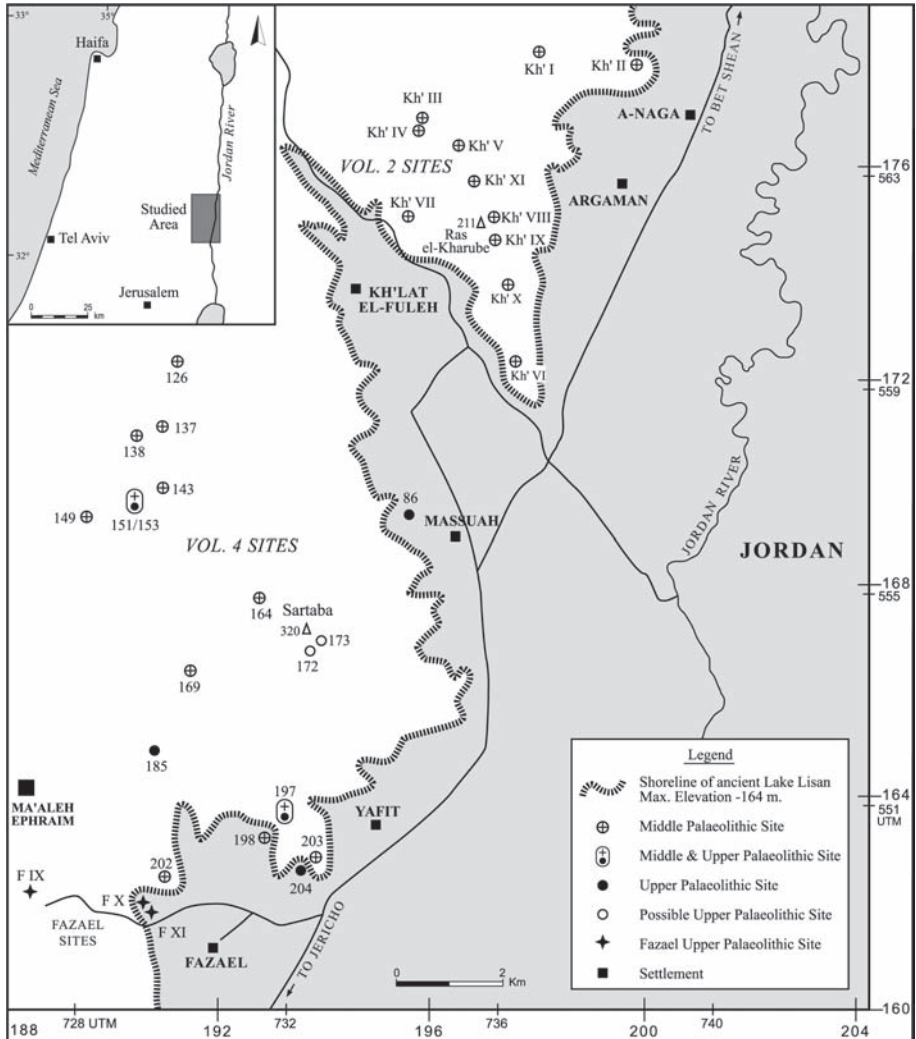
7. The Kebaran – a continuation of the Upper Palaeolithic – started about 20 ka BP at a time when



596. Lake Lisan level (after Bartov et al. 2002)

Lake Lisan still covered a part of the studied area; therefore, some early Kebaran sites could also be covered by lake deposits.

8. The site of Ohalo 2 was later dated to 23 ka BP and attributed to the Upper Palaeolithic.



597. Site location map of Middle and Upper Palaeolithic sites and Lake Lisan shoreline at its highest level – 164 m below sea level.

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APPENDIX B - COIN FINDS

COIN FINDS FROM THE EASTERN VALLEYS AND THE DESERT FRINGES

Ya'akov Yanai

A cataloged description of the coin finds is presented here as an appendix. Most of the coins are identified, and on many of them the names of the rulers, mints and dates are legible.

A period of over 2000 years is embodied in the numismatic finds, from the beginning of the 2nd century BCE until the present day. This appendix presents the origins and date of minting of the coins, some minted "locally" in the Holy Land but most minted throughout the empires that controlled the land during the following periods: Hellenistic, Roman, Byzantine, Moslem (Umayyad, Abasside, Mameluk), and Ottoman; as well as modern coins from Iraq, Jordan and Israel.

Numbers of sites in the general book are used here. Within each site the coins are numbered chronologically. The information about each coin is in the following order:

A. General description:

1. Period;
2. Ruler and years of his reign;
3. Material denomination;

4. Diameter;

5. Weight.

B. Description of obverse and reverse sides, including inscriptions, date and mint.

C. References.

Abbreviations:

AR	–	Silver;
Circular ins.	–	Circular inscription;
Ex	–	Exergue;
Ref.	–	Reference;
gm.	–	grammes;
l.	–	left;
mm.	–	millimetres;
Obv.	–	Obverse;
Rev.	–	Reverse;
r.	–	right;
stg.	–	standing;

I wish to express my special thanks to Adam Zertal, who entrusted me with the task of processing the numismatic finds, to David Shklovski for the photographs and to Ariel Berman who helped both with the bibliography and with the general references for the Arabic coins and especially for numbers 2:1; 33:7; 133:3; 158:1; 194:9-10.

Site No. 1:

Khirbet 'Anahum

1. Byzantine, Mauricius Tiberius, 582-602 C.E., Bronze, 1/2 follis – 20 numia, 21.7-23.5 mm, 5.85 g.

Obv.: Full-face portrait of ruler, wearing crested helmet, clothed, armored, globe with cross in r. hand.

Circ. Ins.: "DN. MA TIBER P..." embossed ring around edge.

Rev.: Large 'K' with cross above, at l.: 'A/N/N/O', at r.: 7-41 (588/9 C.E.), below: I = Constantinople mint G, embossed ring around edge, axis 7.

Ref.: Type as in Roth 135, no. 71, 81.

2. Modern, State of Israel (founded 1948). Bronze-aluminum, 21.5 mm, 4.15 g, bilingual Hebrew-Arabic inscriptions (from 'agorah' series of תש"ך-תשנ"ט).

Obv.: Inside square depression with rounded corners, palm tree with seven date fronds, two bunches of dates, below 'ישראל' in Hebrew and to l., 'Israel' in Arabic.

Rev.: Inside depression as above; large no. 10, below: agurot/תש"ך, minting date 5720 after Creation – 1960 C.E., Axis 12.

Ref.: Coins of the World 1982, 1084, no. 25.



Site No. 2:

Khirbet esh-Sheikh Safiriyan

1. Early Moslem, Abbaside, Almuwattakil, 232-247 of the Hijrah, 847-861 C.E., Bronze, max. 19 mm, 1.93 g, Fils (section), Arabic inscription.

Obv.: First part of the Shuhadah in center: 'LA ALLAH / ALLAH W../ LA SHARIK...' between two cir-

cles, blurred inscr. (verse 33, chapter 9 of the Quran?).

Rev.: Second part of the Shuhada. Center: 'ALLAH / MUHAMED / RASU../ AL...'; Above: 'ALLAH.' Around, between two circles is the date: '235' of the Hijrah, 849 C.E., Axis 8.

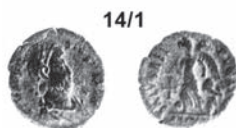
Site No. 14:

el-Khirbeh (ez-Zard)

1. Late Roman, Theodosius I, 379-395 C.E., Bronze, 12.5-13.6 mm, 0.9 g, AE/4, Latin inscription.

Obv.: Ruler to r., adorned with pearl diad., clothed, armored; Circ. Ins.: 'DNTHEODO-SIVSPFAVG'.

Rev.: Victoria dragging captive to l., wreath in hand; below l.: Cross.; Circ. Ins.: 'SALVS REI-PVBLI-CAE'; below mint: 'ANT' - Antioch. Minted between 383-395 C.E., Axis 5.



Ref.: Bruck, 61; RIC IX, 295, no. 70.

2. Bronze, 11.5-12.2 mm, 0.77 g, AE/4, worn. As above [?], ruler and mint unidentified. Minted between 383-395 C.E., Axis 11.

Ref.: Bruck, 61.

3. Bronze, 10.5-11.4 mm, 1.1 g, AE/4, worn, 5th century C.E.

4. Bronze, 9.5-11.1 mm, 0.72 g, AE/4, worn.

Obv.: Ruler to r., clothed. 5th century C.E.

5. Bronze, 9.8-10.4 mm, 0.79 g, minute, AE/4, worn.

Obv.: Ruler to r., clothed. 5th century C.E.

6. Bronze, 10.4-12.5 mm, 0.48 g, unidentified, worn.

7. Early Moslem, AR, 5.5-14 mm, 0.33 g, fragment of unidentified dirham.

8. Lead, 17.5-20 mm, 3.45 g, not definitely a coin.

9. Lead, 11-16 mm, 1.62 g, triangular piece.



Site No. 33:

Khirbet Abu Farhan

1. Late Roman, Valens, 364-378 C.E., Bronze, 16.3-17.5 mm, 1.75 g, AE/3, Latin inscription.

Obv.: Ruler to r., adorned with pearl diad., clothed, armored. Circ. Ins.: 'DNVALENS-PFAVG'.

Rev.: Caesar dragging captive to r., banner in l. hand. Circ. Ins.: 'GLO-RIARO-MANORVM'. Bottom: mint: 'ANT' - Antioch. J, Axis 6.

Ref.: Bruck 35; RIC IX, 274, no. 10(b); 281, no. 35(b).



2. Bronze, 13.3 mm, 0.91 g, AE/4, illegible.

Obv.: Ruler to r., adorned with pearl diadem, clothed, armored.

Rev.: Victoria dragging captive to l., illegible, minted between 383-395 C.E., Axis 6.

Ref.: Bruck 61.

3. Honorius, 393-423 C.E., Bronze, 11-12.2 mm, 1.1 g, AE/4, Latin inscription.

Obv.: Ruler to r., adorned with pearl diadem, clothed, armored. Around: '.....RIVSPF AVG'.

Rev.: Victoria dragging captive at l., worn; bottom: 'CONS' - Constantinople mint, VI. Minted 393-395 C.E., Axis 6.

Ref.: Bruck 61; RIC 9, 236, no. 90(C)

4. Byzantine, Justinian I, 527-565 C.E., Bronze, 29.3-31.1 mm, 14.6 g, Follis-40 Nummia, Latin inscr.

Obv.: Large 'M', cross above it, at l.: seven-pointed star, star at right. Below: I. Bottom: cross and name of mint: 'THEuP' - Theopolis III (Antioch). Minted between 528-538 C.E., Axis 6.

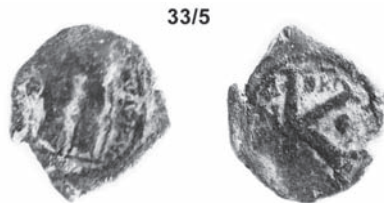
Ref.: Roth 54, no. 273.

5. Justinus II, 565-578 C.E., Bronze, 21-23 mm, 5.28 g, half Follis-20 Nummia, Latin inscr.

Obv.: Justinus II and Sophia sitting on throne, full face; around: '...NVSPPAVG'

Rev.: Large 'K', above 'KC', bottom l.: '.../N/A', r. E=5 (569-570 C. E.), Below: 'TES' (?) - Thessalonica mint, Axis 6.

Ref.: Roth 85, no. 108.



6. Mauricius Tiberius, 582-602 C.E., Bronze, 23-23.6 mm, 6.17 g, half Follis-20 Nummia, Latin inscr.

Obv.: Full-face portrait, wearing helmet with plume, clothed, armored, globe with cross in r. hand, around: 'MAV-TIBERPPA', ring around edge.

Rev.: Large 'K', cross above, at l.: 'A/N/N/O', at right: = 8 = 589/590 C.E., Below: 'E', Constantinople

mint, V, ring around edge, Axis 6.

Ref.: Roth, type as in 134-135.

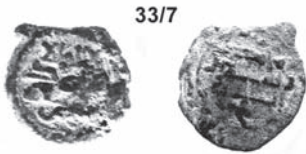


7. Early Moslem, Abbaside, Abu Ja'afar Abdallah Almamun, 196-218 Hijrah, 813-833 C.E., Bronze, 17-19.2 mm, 2.56 g, Follis, domed token, Arabic inscription.

Obv.: Center: 'LA ALLA ILA / ... H / ... K LAHU'. Below – wavy line, rope around edge.

Rev.: In circle: 'MUHAMAD / RASUL / ALLAH'. Circ. Ins. Blurred. Ramlah mint, Axis 1.

Ref.: Lavoix I, no. 1586; Berman 1976, similar type, 39, no. 91.



8. Middle Ages, Mameluk, Abu Bakr, al Mansur Seif ed-Din Abu Bakr, 741-742 Hijrah, 1341 C.E., Bronze, 17-17.6 mm, 3.46 g, Fils, Arabic inscriptions.

Obv.: Star of David, points between its tips, circle around. Illegible date in star.

Rev.: Star of David as above, inside: 'B / DA / MASHK' (minted in Damascus).

Ref.: Balog, Mameluks 164-5, type

as in 267-268.

9. Sh'aban II. el-Ashraf Naser ed-Din Sh'aban II, 764-778 Hijrah, 1363-1377 C.E., Bronze, 17.2-19 mm, 2.95 g, Fils, Arabic inscriptions.

Obv.: Around and among design of 'eye' with flower bulbs on edges: '... TAN / ALMALK', circle at edge.

Rev.: Inside wreath of flower bulbs: illegible name of mint.

Ref.: Balog, Mameluks 221, no. 458.

10. Bronze, 16-17.7 mm, 1.85 g, Fils, worn, illegible.

11. Ottoman Period. Bronze, 16.1-17.1 mm, 8.13 g, Fils, thick token, Arabic inscription.

Obv.: Inscr: 'MISR'. (Name of coin in Egypt: alkahrah – Cairo).

Rev.: Complicated design [net].

Ref.: Type as in Berman 1976, 122, no. 385 and drawing on back cover.

12. Muhammed V, 1327-1336 Hijrah, 1909-1918 C.E., Cupro-nickel, 16 mm, 1.85 g, 5 Farah, Arabic inscription.

Obv.: Monogram (tughrah) of ruler's name; at right: Rashad ('walks the straight path'). Below: 'SIT-TAH' (Year 6 – 1914-1915 C.E.). Around; circle of pearls; above and below: decoration of sheaves at r., decoration of olive branch at left; at edge: circle with short, perpendicular lines.

33/12



Rev.: '5 FARA'. Around, frame of pearls (almost full crescent moon). Above center, between two stars: 'DAWLAH UTHMANIYAH'. R. and l. 'DARAB FI QUSTAN-TINA'. Below center: decoration, '1327' (1327 Hijrah – year of sultan's ascension to throne), Axis 12.

Ref.: Coins of the World 1982,

1749, no. 43.

13. Modern, Iraq, Feisal I, 1921-1933 C.E., Bronze, 19 mm, 2.43 g, Fils, Arabic inscription.

Obv.: Ruler to r., Circ. Ins. 'FAISAL ALAWAL MALK ALIRAQ'.

Rev.: 'FILS' (1). Below: 'Fils' in central circle. Above and below: 'ALMAMLAHAH ALIRAQIYAH'; at r. '1349' (Hijrah), at l.: 1931 (C.E.), Axis 12.

Ref.: Coins of the World 1982, 1065, no. 1.

33/13



Site No. 44:

Khirbet Ibziq (Upper)

1. Roman, Faustina II (wife of Marcus Aurelius), died 176 C.E., Neapolis (Shechem). Bronze, 19.9-22.6 mm, 5.63 g, city mint, Greek inscr..

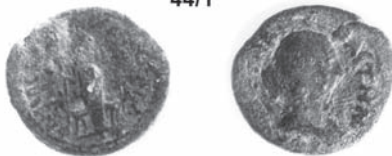
Obv.: Empress to r., hair pulled back in 'ponytail', clothed; circ. Ins (Greek): '...TEINANCEBAC-THN'.

Rev.: Artemis of Ephesus facing front, in her arms, 'shepherd's crocks'. Two deer stg. beside, facing back and up toward goddess. Around: 'FLNEAC ...CIRIAPA';

On two sides of figure in upper area: 'ET – 4' (year 90 of city calendar, 162 C.E.), Axis 12.

Ref.: Rosenberger III, 8, no. 19.

44/1



2. Late Roman, Arcadius, 383-408 C.E., Bronze, 111.4-13 mm, 0.88 g, AE/4, Latin ins.

Obv.: Ruler to r., adorned with pearl diadem, clothed, armored. Around:

'...ARC...'

Rev.: Victoria dragging captive at l., worn. Minted between 383-395 C.E., Axis 7.

Ref.: Bruck 61.

3. Byzantine, Justinian I, 527-565 C.E., Bronze, 27.8-29.5 mm, 11 g, half follis – 20 nummia, Latin inscr.

Obv.: Ruler to front, wearing helmet with plume, clothed, armored. In r. hand: globe with cross, at r.: cross; around: 'DNIVSTIN-IAN-VSPPAVG'.

Rev.: Large 'K', cross above, at left: 'O/N/N/A', at right: XII = 538/9 C.E., Below: B.

Ref.: Roth 36, no. 107.



4. Tiberius II Constantine, 578-582 C.E., Bronze, 28.4-30 mm, 11.5 g, follis – 40 nummia, Latin inscr.

Obv.: Ruler to front, wearing crown with cross, clothed as consul. In r. hand: 'mappa', in l. hand: staff with eagle/vulture, illegible.

Rev.: Large 'M', cross above, at left: 'O/N/N/A', at right: 91/1 = 8 = 581/2 C.E., Below: 'NIKOA' - Nicomedia mint, I, Axis 1.

Ref.: Roth, type as in 113, no. 7.

5. Constans II, 641-668 C.E.,

Bronze, 19.8-25.9 mm, 4.71 g, Follis – 40 Nummia, Latin inscr., chopped-off token.

Obv.: Portrait of ruler, full face, wearing crown with cross, clothed. In r. hand: globe with cross, around: 'IMPER-CONST...'

Rev.: Large 'M'; at left: 'A', Below: 'A', at bottom: 'NEO...', attributed to Constantinople mint, Axis 6.

Ref.: Roth, type as in 266, no. 93, pl XXXI:14.



6. Medieval, Mameluk. 15 mm, 1.66 gm, worn.

7. Bronze, 1.5 g, fragment of ring with carved engraving.

8. Modern, Hashemite Kingdom of Jordan (Hussein, ascension to throne 1952). Cupro-nickel, 30.5 mm, 7.46 g, 50 Fils, bilingual (Arabic-English) inscription.

Obv.: '50', above: '1374' (Hijrah), below: '1955' (C.E.), 'HUSEIN MALIK', in central circle with crown above it. Bottom, l. and r.: wreath of date cluster and two sheaths. Next to crown: flower with seven petals; around: at r. below and at l. inscription: 'ALMAMLAHAH ALARABIYAH ALHASHEMIYAH',

Rev.: '50' in inner circle, and above:

‘FIFTY FILS’, below: ‘1955’ and seven-pointed star; around: ‘THE HASHEMITE KINGDOM OF JORDAN’, Axis 12.

Ref.: Coins of the World 1982, 1174, no. 11.



44/8



Site 114:

Khirbet Umm el-‘Uqbeh (A)

1. Late Roman, Constantine II 317-337 C.E., Caesar during his father’s reign (337-340 C.E. – Augustus) Bronze, 18.5 mm, 2.65 g, AE/3, Latin inscription.

Obv.: Laur. Head r., ins: ‘CONSTANTINVSIVNNOBC’.

Rev.: Two soldiers stg. either side of two standards. Circ.ins.: ‘GLORIAEXERC-ITVS’; In ex.: ‘ASIS’ - Siscia mint, A, minted between 330-335 C.E., Axis 12.

Ref.: RIC VI, 453 no. 220.



2. Theodosius I (?), 379-395 C.E., Bronze, 12.8-13.5 mm, 0.8 g, AE/4, worn.

Ref.: Bruck, 92-93.

3. Arcadius, 383-408 C.E., Bronze, 12.1-12.9 mm, 0.73 g, AE/4, illegible.

Obv.: Ruler to r.

Rev.: Victoria dragging captive to left. Minted between 383-395 C.E., Axis 5.

Ref.: Bruck 61.

4. Bronze, 10 mm, 0.90 g, Minima, worn. 5th Century C.E.

5. Bronze, 10 mm, 1.15 g, Minima, worn. 5th Century C.E.

6. Bronze, 11 mm, 0.58 g, fragment of AE/4, worn. 5th Century C.E.

7. Seventh Century C.E., transition between Byzantine and Moslem Periods. Bronze, 14-16.5 mm, 0.98 g, worn, cut from sheet of bronze.

Obv.: Ruler, facing front.

Rev.: Large ‘M’, underlined, Axis 11.

Ref.: Kirkbride 1948, 62, nos. 57-58; Qedar 1988/89, 30, series A-E (general).

8. Medieval, Ayyubides (Zengis?), El-Malek el-‘Adel Nur ed-din Mahmud (?), 541-569 Hijrah,

1146-1174 C.E., Bronze, 18.5-21.5 mm, 3.98 g, Fils, Arabic inscr.

Obv.: Unclear, plant adornment.

Rev.: Unclear inscription (Damascus mint?), plant adornment.

Ref.: Berman 1976, type as in 63,

no. 163.

9. Medieval, Mameluk? Bronze, 11.2-14.7 mm, 0.82 g, illegible.

10. Bronze, max. 14 mm, 0.46 g, illegible.



Site 127:

Khirbet Musheibik

1. Medieval, Artikim, Mardin Najm ed-Din Alfi, 547-572 Hijrah, 1152-1176 C.E., Bronze, 32.8 mm, 9.53 g, 'Dirham AE', Arabic inscription.

Obv.: Two images facing each other, each adorned with diad (borrowed from type of Imperial Greek minting of Asia Minor from 1st Century C.E.); above: 'NAJM ALDIN'; below: 'MALK DIAR BAKR', tribal sign.

Rev.: Two stg. figures, facing for-

ward; head of figure on right encircled by halo, crowning figure on left (borrowed from type of Byzantine minting from middle 10th to 12th Century C.E.). Inscr. around and between figures: 'ABU ALMUD-HAFR ALFI BIN TAMARTASH IL GHAZI BIN ARTAQ', Axis 4; Two perforations for hanging (one broken, not in use), matching heads of coin on either side.

Ref.: Berman 1976, 58, no. 147; Lane-Poole III, 142, no. 372.

127/1



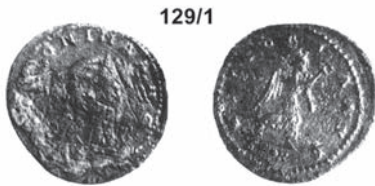
Site 129:

Khirbet es-Samrah

1. Roman, Salonina (wife of Gallienus). Silver (billon), 21.1-22.4 mm, 2.77 g, Antoninianus, Latin inscription.

Obv.: Ruler facing r., on crescent, adorned with diadem, clothed. Around: 'SALONINAAVG'. Pearls around edge.

Rev.: Victoria marching at r., furling banner in r. hand, holding bunch of dates on l. shoulder; around: 'VICTORIAAVG' under: 'SI B' - Siscia-B min. Combination of Salonina and Gallienus coins. Minted between 260-268 C.E., Axis 12.



Ref.: *Obv.* only: RIC V:1, 198-199 (general), Only *Rev.*: like type of Gallienus – RIC V:1, 183, no. 588.

2. Bronze, 13.3 mm, 1.3 g, local minting, worn.

3. Late Roman. Bronze, 11.4-12.5 mm, 0.84 g, AE4, Latin inscription.

Obv.: Ruler facing r., adorned with pearl diad, clothed, illegible.

Rev.: Victoria dragging captive l., also holding booty in r. hand. Around: 'SALVSREI-PVBLICAE' ; bottom: 'SMR' - Rome mint:

ROMA. Minted between 388-395 C.E., Axis 6.

Ref.: Bruck 61; RIC IX, type as in 133-134, no. 64; 136, no. 69 (general and without Christogram).

4. Bronze, 11.6-12.7 mm, 1.35 g, AE/4, worn, last quarter of 4th Century C.E.

5. Bronze, 11.7 mm, 1.13 g, worn.

6. Bronze, 10.3 mm, 1.0 g, worn, 5th Century C.E.

7. Bronze, 7.1 mm, 0.52 g, Minima.

Obv.: Worn. *Rev.*: Illegible, possibly monogram, 5th Century C.E.

8. Bronze, 7.7 mm, 0.63 g, Minima, worn, 5th Century C.E.

9. Mameluke, en-Nasr Aladdin Muhammed?, third reign, 709-741 Hijrah, 1310-1341 C.E., Bronze, 14.6-16.5 mm, 1.40 g, Fils, Arabic inscription.

Obv.: Worn, 'MUHAMAD..', frame adornment.

Rev.: Worn.

Ref.: Balog, Mameluks (general), 152, no. 230b, Table IX; 158, no. 248, Table X.

10. Bronze, 15.8-17 mm, 0.53 g, Fils, worn, undated.

11. Bronze, 19.4-23.1 mm (inner-outer), 2.0 g, ring, undated.

Site 133:

Khirbet Umm el-Qatan

1. Medieval, Ayyubids, el-Malek el-Kamel I en-Nasr ed-Din Moham-med, 615-635 Hijrah, 1218-1238 C.E., with Caliph el-Mustanaser, 623-640 Hijrah, 1226-1242 C.E., Bronze, 21.5 mm, 3.18 g, Fils, Arabic inscription.

Obv.: partially erased inscription: '...ALMALK ALKAMEL'.

Rev.: as above: 'ALIMAM'.

2. el-Kamel I, as above. Bronze, 17-19 mm, 2.85 g, Fils, Arabic inscription.

Obv.: inside square frame with external dots: '...K ALKA .../ MIN/...ALMALK....'.

Rev.: '...MAD BIN ABI / B...DIN ABU...', Axis 3. Attributed to Damascus mint.

Ref.: Balog, Ayyubids, no. 465, completion of reading from Beth

Shean excavation.



3. Bronze, 21.5 mm, 1.95 g, Fils, disfigured by knife.

4. Mameluks, ez-Zahr Rukkan ed-Din Baibars I, 658-676 Hijrah, 1260-1277 C.E., Bronze, 18.5 mm, 2.22 g, Fils, Arabic inscription.

Obv.: Indistinct inscription; lion walking, facing l.

Rev.: Indistinct inscription.

Ref.: Balog, Mameluks, 103 (general).

5. Steel arrowhead, overall length 60 mm, 5.93 g, triangular blade 20 mm long, max. width of side 6.8 mm, point 25 mm long.



Site 143:

Khirbet Humsah

1. Roman, Aurelianus, 270-275 C.E., Bronze coated with silver, 21-22 mm, 2.78 g, Antoninianus, Latin inscription.

Obv.: Ruler to r., adorned with radial crown, armored, long, thin neck; around: 'IMPCAVRELIAN-

VSAVG'.

Rev.: Female figure at l. presenting wreath to ruler stg facing her at r.; seven-pointed star between them. Around: 'RESTITVTOR-BIS', below: 'KA' -Tripoli mint, Phoenicia, 3rd Period, Axis 12.

Ref.: RIC V, 309, no. 389.



2. Late Roman, Constantine I (the Great), 307-337 C.E., Bronze, 19.6-20.7 mm, 3.63 g, reduced Follis, Latin inscription.

Obv.: Ruler r., adorned with wreath, clothed, armored, around: 'IMCONSTANTINVS PFAVG'.

Rev.: Sol stg naked at l, adorned with radial crown, r. hand open, globe in l. hand, cloak around neck and over l. shoulder, at l. seven-pointed star; around: 'SOLI INVIC TOCOMITI'; bottom: 'PT' - Ticinum mint. Minted between 313-318 C.E., Axis 6.

Ref.: RIC VI, 361, no. 14.



3. Constantius Gallus, 351-354 C.E. (under Constantius II). Bronze, 21-23 mm, 3.38 g, Centenionalis, Latin inscr.

Obv.: Ruler facing r., bareheaded, clothed, armored, the letter Delta to l. of back of neck; around: 'DN.....STANTI-VSNOBCAES'.

Rev.: Legionnaire spearing fallen horseman at l.; around: '...MPREPARATIO'; below: 'ALEA' - Alexandria mint, A, Axis 6.

Ref.: Bruck 17-20.



4. Early Moslem, transition between Byzantine and Moslem. Bronze, 14.5-16.5 mm, 2.53 g, Follis cut rectangularly; Arabic inscription?

Obv.: figure facing forward.

Rev.: Large 'm', at left: 'TAYB', (טב = TOV?* or Tiberias?); below: horizontal line; Axis 11.

* possibly Countermark: 'TOV', Qedar, 1988/9, 32; and with translation of Meshorer 1965/6, 32-34.



Site 158:

Khirbet el-Khreibat

1. Medieval, Ayyubids, el-Malek en-Naser Salah ed-Din Yusef ibn Ayyub, 564-589 Hijrah, 1169-1193 C.E., Bronze, 23-24.8 mm, 5.8 g, Fils, Arabic inscription, adorned letters.

Obv.: Within inner ring of pearls: 'ALMALK ALNASR'; above: lily; below: plant adornment, around outside 'SULTAN ALMUSLI-MIN'; pearls around edge.

Rev.: Within inner ring of pearls: 'YUSEF / BIN AYYUB'; above, decoration resembling flat letter 'V'; below: six-pointed star; around outside: 'DARB BIDIMASHK SANA SIT WATAMANIN', (5[86] Hijrah, 1189/90 C.E.), Axis 3.

Ref.: Berman 1976, 68, no. 176; Balog, Ayyubids, no. 148.



2. El-Malek et-Thahr Gyat ed-Din

Ghazi, 582-613 Hijrah, 1186-1216 C.E., with Caliph en-Nasr, 575-622 Hijrah, 1180-1225 C.E., Bronze, max. 22.2 mm (fragment), 2.81 g, Fils, Arabic inscr.

Obv.: 'ALMALK ALTHAHR...', inside octagonal star of pearls, bordered by line. Around: 'AHAD ASHAR WASI...' (6[11] Hijrah – 1213/14 C.E.). Pearls around edge.

Rev.: 'ALIMAM ALNASR...'; inside star as above; around: 'MAD RASUL ALLAH'. Pearls around edge, Aleppo mint, Axis 1.

Ref.: Berman 1976, type as in 76, no. 206, but date is different and name of minter is outside token; Balog, Ayyubids, no 670.



3. Bronze, max. 15 mm, 0.37 g, broken coin, worn.

Site 168:

Khirbet Bet Farr (B)

1. Late Roman, Constantine I (the Great), 307-337 C.E., Bronze, 16.5-17.5 mm, 2.07 g, AE3, Latin inscr.

Obv.: Ruler facing r., adorned with diadem with rosette, clothed, armored. Around: 'CONSTANTINVS MAX AVG'; pearls around edge.

Rev.: Two soldiers stg. with two banners between them, spear in outer hand of each, inner hand on shield. Around: 'GLORIA EXERCITVS'. At bottom: mint – illegible. Minted between 330-335 C.E., Axis 6.

Ref.: Bruck 25-28.

2. Arcadius, 383-408 C.E., Bronze, 13-14 mm, 1.31 g, AE/4, Latin inscription.

Obv.: Ruler to r., adorned with pearl diadem, clothed, armored, around: 'DN ARCADIVS PF AVG'.

Rev.: Victoria dragging captive to l., carrying booty on shoulder and in r. hand; at l.: Christogram. Around: 'SALVS REI PVBLICAE'; below: 'CONA' - Constantinople mint, A, between 388-392 C.E., Axis 12.

Ref.: RIC IX, 234, no. 86(c).

3. Byzantine, Heraclius or Constans II?, first half of 7th Century C.E., Bronze, max. 16 mm, 1.44 g, Follis – 40 Nummia (fragment). *Obv.*

and *rev.* indistinct.

4. Transition Byzantine-Arabic. Bronze, 17.5-24.5 mm, 3.30 g, Follis – 40 Nummia.

Obv.: Ruler standing front, wearing crown, clothed, scepter topped by cross in right hand, globe with cross in l. hand, illegible.

Rev.: Large 'm', horizontal line connecting, at center, three parts of letter; above: diagonal cross (perhaps X?); below: horizontal line, illegible, Axis 12.

Ref.: Kirkbride 1948, general, no exact parallel.



5. Medieval, Mameluke, ez-Zahr Rukkan ed-Din Baibars I, 658-676 Hijrah, 1260-1277 C.E., Bronze, 14.5-18.0 mm, 2.85 g, Fils, cut token, Arabic inscr.

Obv.: Lion moving l., tail raised; at left and below: lines of frame; attributed to Damascus mint, Axis 8.

Rev.: esh-Shuhadah: 'LA ALLAH IL / MUHAMAD'.

Ref.: Balog, Mameluks (general), type as in p. 105, Table IV: 98-100.

168/5



6. En-Nasr Nasr ed-Din Muhammed, Third Reign, 709-741 Hijrah, 1310-1341 C.E., Bronze, 17.0-19.5 mm, 2.71 g, Fils, Arabic inscriptions.

Obv.: 'ALMALK ALNASR...'; Above and below: floral decorations; around edge: circle and dots outside it.

Rev.: In center, round 'sign' with five-petalled flower sunk inside; around and inward: 'DARB BIDI-MASHQ SANA TALATIN' - 7(30) Hijrah [1329 C.E.].

Ref.: Balog, Mameluks, 162, no. 261; Berman 1976, 88, no. 22.

7. Bronze, 17-20 mm, 1.77 g, Fils, Arabic inscriptions.

Obv.: 'ALMALK ALNASR...' around circle, and illegible inscription.

Rev.: 'MIN / ALMALK / ...' around circle, and illegible inscr. First half of 14th Century C.E., Axis 10.

Ref.: Balog, Mameluks, type similar to 148, no 215.

8. Al-Ashraf Nasr ed-Din Sh'aban II, 764-778 Hijrah, 1363-1377 C.E., Bronze, 14.5-16.0 mm, 0.74 g, thin, Arabic inscriptions.

Obv.: '...K/ ALAF/SHAR...', underlined.

Rev.: 'HAD SABU.../ SANA...'

- 7(71) Hijrah [1369/70 C.E.], attributed to Damascus mint.

Ref.: Balog, Mameluks (general, due to poor preservation) 220, no. 456.

9. Bronze, 16.5-17.5 mm, 1.99 g, Fils, Arabic inscriptions.

Obv.: 'ALMALK ALAF...'; above: dividing line and decoration.

Rev.: 'SHU...'N; below: dividing line and inscription: '...SHAF...'; the coin has a hole in it, Axis 9.

Ref.: Balog, Mameluks (general) similar to type on 220, Table XVII: 454.

10. Mameluk, undefined. Bronze, 19.5-21.5 mm, 3.29 g, Fils.

Obv and Rev.: Indistinct Star of David, 14th-15th Century C.E.

11. Bronze, 15 mm, 0.56 g, thin, undefined.

Obv.: 'ALSULTAN ALMALK A...'

Rev.: Inner circle, worn, 14th-15th Century C.E.

12. Bronze, 13.5-15 mm, 0.59 g, thin, undefined.

Obv.: At r., frame of dots: '...AL-MALK...', worn.

Rev.: In frame of dots: 'HAD...', worn, 13th-15th Century C.E.

13. Bronze, 14-16 mm, 0.80 g, thin, undefined.

Obv.: 'ALNASR...', worn.

Rev.: Above frame of dots, at edge: '...MD...', worn, 13th-15th Century

C.E.

14. Bronze, 14-15 mm, 0.54 g, thin, undefined.

Obv and Rev.: worn, illegible.

15. Bronze, 14 mm, 0.40 g, thin,

not definitely a coin, worn.

16. Bronze, max. 21 mm, 0.67 g, metal sheeting (cut), not definitely a coin, indistinct. Ottoman.

17. Lead, weight, diameter 14.5 mm, height 20.5 mm, 24.34 g.



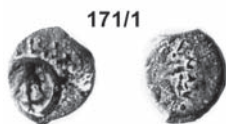
Site No. 171:

Tel Miski

1. Hasmonaean (Late Hellenistic), Alexander Yannaeus, 103-76 B.C.E., Bronze, 12-13.5 mm, 0.97 g, Prutah, Greek-Aramaic bilingual inscription, excentric minting.

Obv.: Anchor in circle; around: '...YBASΣ...'.
Rev.: Eight-pointed star in dotted circle, around: 'מלכא אל' (King Al[exander]), Axis 12.

Ref.: Meshorer 1982 A, 79-80, type Cd, Table 7.



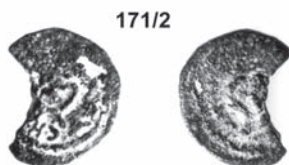
2. House of Herod (Early Roman), Herod the Great, 37-4 B.C.E., Bronze, max. 19.1 mm, 2.91 g, double Prutah, Greek inscription (partially broken).

Obv.: X surrounded by ribbon; around: 'HP... AECY', pearl edge.

Rev.: Three-legged table on horizontal line, at left and right date

fronds, pearl edge, Axis 12.

Ref.: Meshorer 1982 A, 24-25, 236; no. 7-8, Table 7.



3. Late Roman. Constantius II, 337-361 C.E., Bronze, 15.3-16.4 mm, 1.99 g, AE/3, Latin inscription.

Obv.: Ruler to r., adorned with diadem, clothed, armored, around: '..TIVSPFAVG'.

Rev.: Legionnaire spearing fallen horseman. Around: '..EMP-REPARATIO'. In area at l.: 'M'; below: 'SMTHES' – Thessalonica mint, minted between 354-361 C.E., Axis 12.

Ref.: Bruck 20.



4. Eudochia (wife of Arcadius, 395 C.E., died 404 C.E.). Bronze, 12.3-13 mm, 1.16 g, AE/4.

Obv.: Ruler to r., adorned with elegant diadem, dressed, illegible.

Rev.: Seated figure facing forward, blurred, Axis 12.

Ref.: Bruck 42.

5. Theodosius II, 402-450 C.E., Bronze, 12.4-13 mm, 1.37 g, AE/4,

Latin inscr.

Obv.: Ruler to r., adorned with diadem, dressed, star at left. Around: 'DNTHEO...'

Rev.: Theodosius II stg. between Arcadius and Honorius facing forward, holding spears, illegible. Axis 12.

Ref.: Bruck 42; Kent and Carson 1965, Table III: 2214.



Site No. 174:

Tel el-Hadad

1. City coin, Neapolis (Shechem), Elagabalus, 218-222 C.E., Bronze, 23.5-24.8 mm, 12.35 g, Greek inscr.

Obv.: Ruler to r., adorned with wreath (?); around: 'ACT...OC', pearl edge, signs of sawing on token.

Rev.: Mount Garizim with temple and altar, with steps leading to them from avenue of columns at foot of mountain; around: '...A...TTOA...'; Axis 12.

Ref.: Rosenberger III, 11, no. 37-39.

2. Late Roman. Bronze, 13-14.4 mm, 0.74 g, AE/4.

Obv.: Ruler to l., adorned with wreath, clothed, indistinct inscription.

Rev.: Inside diadem: 'V/VOT', in-

distinct. Last quarter of 4th Century C.E., Axis 12.

Ref.: Bruck 94.

3. Byzantine. Heraclius, 610-641 C.E., Bronze, 30.6-32.4 mm, 10.90 g, Follis – 40 Nummia, stamped over older mint, Latin ins.

Obv.: Heraclius at l., Heraclius Constantinus at r., stg. and facing forward, wearing crowns, dressed, each holding globe with cross in right hand; around, distorted inscription: '...DDNhERA...CONS'.

Rev.: Large 'm'; above: cross; at l.: 'N/N/A/O'; at r.: 41 = 6 (615/6 C.E.); below in space: 'E'; at bottom: 'CO(N)' - Constantinople mint, V, Axis 7.

Ref.: Sear 1974a, 151, no. 805; Roth (general), type as in 201 no 139 (different date).

Site No. 176:

Khirbet Tel el-Fukhar

1. Roman, Roman consuls in Judea, Festus, 59-62 C.E., under reign of Nero (54-68 C.E.).

Bronze, 16.8-18.4 mm, 2.73 g, Prutah, Greek inscription.

Obv.: Name of emperor inside wreath: 'NEP/WNO/C' pearl edge.

Rev.: Date frond; around: 'LEKA-ICAPOC', year 5 of Nero's reign (59 C.E.), Axis 12.

Ref.: Meshorer 1982, B, 183, 285; no. 35, Table 33.



2. City coin, Caesarea, Hadrian, 117-138 C.E., Bronze, 18.2-20 mm, 6.40 g, Latin inscription.

Obv.: Ruler to r., adorned with wreath, clothed, armored. Around: 'IMPTRAHA DRIANOC'AA'; pearl edge.

Rev.: Apollo stg naked, facing left, snake in right hand, left hand leaning on tripod for incense, cloak draped over neck and l. shoulder; around: 'CIFA VG-CAESAR'; pearl edge, Axis 12.

Ref.: Kadman 1957, 102, Table III: 29; Rosenberger, II, 4, no. 24.

176/2



3. City coin, Neapolis (Shechem). Bronze (Orichalcum), 23.8-24.3 mm, 9.6 g, Greek inscr.

Obv.: Ruler to r., rectangular countermark with head (?) on neck at right; around: '...AV...KAICAP', worn.

Rev.: Stg. figure (resembles Zeus Heliopolitas) facing front, worn, Axis 12.

4. Late Roman, Probus, 276-282 C.E. Silver-coated bronze (remnants), 21.1-22.2 mm, 4.15 g, Antoninianus, Latin inscription, pearl edge on both sides of coin.

Obv.: Ruler to r., adorned with radial crown, clothed, armored. Around: 'IMPCMAVRPROBVS AVG'.

Rev.: Probus clothed, stg. and facing l., holding scepter with eagle at top, accepting globe from Jupiter at r., naked except for cloak draped from neck onto left shoulder; long scepter in l. hand. Around: 'CLEMENTIAT-EMP'; below in space: 'E'; at bottom: 'XXI'; attributed to Siscia mint. Axis 6.

Ref.: RIC V: 2, 86, no. 643.



5. Carus, 282-283 C.E. Silver-coated bronze (remnants), 20.5-22.2 mm, 3.28 g, Antoninianus, Latin inscr, pearl edge on both sides of coin.

Obv.: Ruler to r. adorned with crown of rays, clothed, armored. Around: 'IMPCMAVRCARVSPFAVG'.

Rev.: Carus, clothed, stg. at l. with dagger, receiving Victoria presenting him with wreath, from Jupiter stg. at r., clothed, long scepter in l. hand, star above; around: VIRT-VSAV-GGG'; below in space: 'A'; at bottom: 'XXI', attributed to Antioch mint, Axis 12.

Ref.: RIC V:2, 150, no. 125.

6. Constantine I (the Great), 307-337 C.E., Bronze, 16-18 mm, 2.57 g, AE/3, Latin inscr, pearl edge on both sides of coin.

Obv.: Ruler to r., adorned with embedded diadem, clothed, armored. Around: 'CONSTANTI-NVS-MAXAVG'.

Rev.: Two soldiers stg with two banners between them, spear in outside hands, inside hands on shield; Around: 'GLOR-IAEXERC-ITVS'; below: 'SMANB' - Antioch mint, II, minted between 330-335 C.E., Axis 6.

Ref.: RIC VI, 693, no 86; Bruck 25.



7. Constantius II, 337-361 C.E., Bronze, 17.3-18 mm, 2.07 g, AE/3, Latin inscriptions.

Obv.: Ruler to r., adorned with diadem, clothed, armored; around: 'DNCONSTANTIVSPFAVG'.

Rev.: Legionnaire stabbing fallen horseman; around: 'FELTEMP...RATIO'; at bottom: 'ALES'; Alexandria mint, V, minted between 350-361 C.E., Axis 12.

Ref.: Bruck 20-21.

8. Bronze, 14.7-15.8 mm, 2.18 g, AE 3/4, Latin inscr.

Obv.: Ruler to r., adorned with diadem, clothed, armored, illegible.

Rev.: Legionnaire stabbing fallen horseman; around: '...TEMP-REPARATIO', minted between 354-361 C.E., Axis 10.

Ref.: Bruck 20-21.

9. Bronze, 14.7-15.7 mm, 2.32g, AE 3/4, Latin inscr..

Obv.: Ruler to r., adorned with diadem; around: 'DNCON...'.

Rev.: Legionnaire stabbing fallen horseman; around: '...REPARATIO', minted between 354-361 C.E., Axis 10.

Ref.: Bruck 20-21.

10. Theodosius I, 379-395 C.E.

13.6-14.7 mm, 1.28 g, AE/4, Latin inscr.

Obv.: Ruler to r., adorned with pearl diadem, clothed, armored; around: 'DNTHEODO-SIVSPFAVG'.

Rev.: Victoria dragging captive to l.; carrying booty on her shoulder, held in r. hand; at l.: Christogram. Around: 'SALVSREI-PV-BLICAIE'; at bottom: 'SMKA' - Cyzicus mint, I/A, minted between 388-395 C.E., Axis 6.

Ref.: RIC IX, 246-247, no. 26(a), 30(a).

11. Honorius, 393-428 C.E. 12-12.5 mm, 0.77 g, AE/4, Latin inscr.

Obv.: Ruler to r., adorned with diadem, clothed, armored; around: '...RIVSPFAVG'.

Rev.: Victoria dragging captive leftward; carrying booty on her shoulder, held in r. hand; at l.: Christogram. Around: '...PVBLICAE'; at bottom: 'ANS' - Antioch mint, V/6, minted between 393-395 C.E., Axis 12.

Ref.: RIC, IX, 295, no. 70(c).

12. Byzantine, Justinus I, 518-527 C.E., Bronze, 27.9-31.6 mm, 16.40 g, Follis - 40 Nummia, Latin inscr.

Obv.: Inside round depression, portrait of ruler to r., adorned with diadem, clothed, armored; around: '...-NVSPPAVG'.

Rev.: Large 'M', cross above, six-

pointed star at left, cross at r. Below: Δ; at bottom: 'CON' - Constantinople mint, IV, relief hoop around, Axis 6.

Ref.: Roth 14, no. 32.

13. Justinianus I, 527-565 C.E.,

176/12



Bronze, 28.6-35.2 mm, 15.79 g, Follis - 40 Nummia, Latin inscr. Dotted circle at edge on both sides of coin.

Obv.: Ruler to r., adorned with diadem, clothed, armored; around: 'DN.-ANPPAV'.

Rev.: Large 'M'; above; cross; at l. and r.: eight-pointed star. Below: Δ; at bottom: '...TIX', Antioch mint, IV, minted between 527-528 C.E., Axis 5.

Ref.: Roth, type as in 53, no. 265, Table VIII:1.

176/13



14. Bronze, 14.7-15.6 mm, 2.62 g, 12 Nummia, Latin-Greek bilingual inscriptions.

Obv.: Ruler to r., remnants of inscription: 'DNIV...'.
Rev.: Large 'I+B'; above: cross; at bottom: 'AΛEZ' - Alexandria mint, Axis 7, could belong to three different rulers between 518-578 C.E.

15. Early Moselm (Umayyad), post-Reform – 697-750 C.E., Bronze, 16.3-17.5 mm, 3.72 g, Fils, Arabic inscription within circle. Obv and

Rev.: Shuhadah, 'LA ALLAH / ILA ALLAH / WAHDAH MUHAMAD / RASUL/ ALLAH', Axis 10.

Ref.: Walker, type as in 210, no. 638, Table XXIII.



16. Bronze, 19.5-21.5 mm, 3.32 g, Fils, Arabic inscription.

Obv.: Inside three concentric circles, first part of Shuhadah: 'LA ALAH / ILA ALLAH / WAHDAH ...';

Rev.: Center, other part of Shu-

hadah: 'MUHAMAD / RASUL / ALLAH', above: 'bud'; around: 'BISM ALLAH DARB HADA ALFLUS ...'; mint illegible, eight-pointed star, pearl edge, Axis 3. The three above circles point to mint in Southern Syria-Palestine.

Ref.: Qedar 1988/89, 35.



17. Roman-Byzantine. Bronze (coated?) with silver, 1.43 g; polished fragment of one side of mirror? Radius indicates diameter of 10-14 cm.

18. Modern. Bronze, ball-shaped bell, 22 mm, 33 mm, 16.96 g, decorated with dotted line. Used mostly for herds of sheep.

19. Ottoman, 20th Century C.E. (Mohammed V). Copper, casing of rifle bullet, 7.92 mm, 12 mm, length 53 mm, weight (uncleaned) 11.50 g. The symbols around: '1332' (Hijrah = 1913/1914 C.E.). Unfired.



Site No. 186:

Khirbet el-'Athmawiyyeh

1. Late Roman-Byzantine. Bronze, undefined coin.
 max. 11 mm, 1.17 g, fragment of

Site No. 194:

Khirbet Basaliyyeh

1. Late Roman, Bronze, 11.6-12.5 mm, 0.77 g, AE/4, Latin inscr.

Obv.: Ruler to r., clothed. Around: 'DN....-...AVG', indistinct.

Rev.: Victoria dragging captive toward l.; at l., indistinct cross. Minted between 383-395 C.E., Axis 12.

Ref.: Bruck 61.

2. Byzantine, Justinus II, 565-578 C.E., Bronze, 28.5-31.5 mm, 12.24 g, Follis – 40 Nummia, Latin inscr, circle around edges.

Obv.: Justinus II and Sophia sitting on throne facing front, clothed, illegible.

Rev.: Large 'M'; above: cross; at l.: 'O/N/N/A'; at r.: 'X' (= 574/5 C.E.); below: 'A'; at bottom: 'NIKO' – Nicomedia, mint A [I], Axis 6.

Ref.: Roth 88, no. 149.

3. Bronze, 20-22 mm, 5.49 g, half Follis – 20 Nummia.

Obv.: Justinus II and Sophia sitting on throne facing front, clothed, illegible.

Rev.: Large 'K'; above: cross; at l.: 'O/N/N/A'; at r.: 'Δ' (= 568/9 C.E.); below: 'TES' – Thessalonica mint, Axis 6.

Ref.: Roth 84, no. 105.

4. Heraclius, 610-641 C.E. 28-31.5 mm, 8.99 g, Follis – 40 Nummia, repeat minting, damaged by corro-

sion.

Obv.: Heraclius and Heraclius Constantine stg to front.

Rev.: Large 'M'; above: cross; at l.: 'O/N/N/A'; at r.: large 'X', remnant of former mint; below: "CON" – Constantinople mint, Axis 7.

Ref.: Roth, type as in 201, nos. 139-141.

5. Early Moslem, Byzantine-Arabic transition. Bronze, 15.3-24.6 mm, 2.41 g, Follis – 40 Nummia, illegible.

Obv.: Figure facing front, crowned, clothed, long cross in r. hand.

Rev.: Large 'M', below: horizontal line, Axis 6.

Ref.: Kirkbride 1948 (general).

6. Damascus. 21-22 mm, 4.35 g, Follis-40 Nummia, bilingual Arabic-Greek ins., encircled by dotted line.

Obv.: Figure stg to front, crowned, long cross in r. hand, globe with cross in l. hand. At left: palm frond with 'T' above it.

Rev.: Large 'M'; above: cross; at l.: 'ΔAM'; at r.: '...MUHAMAD', below: crescent; at bottom: 'BIDI-MISHK'; Damascus mint, Axis 3.

No exact parallel; mint details appear on various other coins; see Walker: palm frond – p. 6 no. 9.

7. Medieval, Umaiyyad, post-reform of Abd el-Malk (77 Hijrah, 696/7 C.E.). Bronze, 18-26.5 mm, 2.58 g, Fils, Arabic inscription, repeat minting.

Obv and Rev.: The Shuhadah; in circle: 'LA ALAH / ILA ALLAH / WAHDAH MUHAMAD / RASUL / ALLAH'. indistinct remnants of earlier minting outside circle; Axis 12.

Ref.: Walker, type as in 208, no. 618; Qedar, 1984/5, type as in "group" 3 without mint name, p. 71 no. 56, pl. 21.

8. 712-751 CE, Ramlah mint. Bronze, 13-14 mm, 3.18 g, Fils, Arabic inscription, repeat minting.

Obv.: Within circle of pearls, floral symbol (similar to palm branch); around, part of inscription.

Rev.: Within circle of pearls, crescent; around, part of inscription and part of outer circle; Axis 3.

Ref.: Walker, similar type, 257 nos. 855-879.

9. Medieval, Ayyubids, all-Malk el-'Adel I Seif ed-Din Abu Bakr, 592-615 Hijrah, 1196-1218 C.E.

Bronze, 22.5-24 mm, 6.69 g, Fils, Arabic inscription, worn, cut.

Obv.: Inside dotted circle, five-pointed star: 'AL .../ ALMALK ALA.../; around on outside: 'SAT-MA...BIDIMISHQ'

Rev.: Remnants of inscription: '... ABU BAKR...'; below, complex net

design; Axis 5.

Ref.: Berman 1976, type as in 72, no. 190; Balog, Ayyubids, no. 320.

10. al-Malk el-Mansur Nasr ed-Din Muhammed I, prince of Hama, 587-617 Hijrah, 1191-1220 C.E., under ruler of Aleppo al-Malk al-Thahr Giyath ed-Din Ghazi, 582-612 Hijrah, 1186-1216 C.E., with the Caliph el-Nasr, 575-622 Hijrah, 1180-1225 C.E., Bronze, 20 mm, 2.66 g, Fils, Arabic inscr.

Obv.: 'ALMALK ALTHAHR / / ALMALK ALMANSUR'; above, in center and below: floral design.

Rev.: 'DARB / ALIMAM ALNA.../ BID MAH', circle in margin, dots around it; Axis 4.

Ref.: Balog, Ayyubids, no. 825.



Non-coin metal objects:

11. Byzantine. Bronze weight, 26.06 g, 25.5 mm, 0.9 mm thick, in shape of backgammon playing piece.

Upper side: Round depression, 4.4 mm knob in center; above: cross; at left: 'T' and small circle; at r.: 'A' (one ounce); below, six rays around point. The signs are depressed.

Lower side: Unsuccessful round depression with central point; shallow, engraved scratches in field (possibly monograms?). Encircling side: Two

engraved coils.



Site No. 239:

Khirbet es-Suwedeh

1. Roman, City coin, Hadrian, 130-138 C.E., Aelia Capitolina (Roman Jerusalem). Bronze, 22-23 mm, 8.9 g, Latin inscription, pearl edge.

Obv.: Ruler to r., clothed, bearded, adorned by wreath. Around: 'HADRIANO AVG IMP CAE-TRAIA'.

Rev.: Head of Aelius to r., clothed, bearded. Around: 'L.AELIVS CAE-SA-COL AEL CAP'.

Ref.: Kadman 1956, 80 and no. 8; Meshorer 1989, 23 and no. 8.

2. Caesarea Maritima. Bronze, 17-19 mm, 6.2 g, Latin inscription.

Obv.: Caesar to r., adorned by wreath, clothed. Around: 'IMP-TRAHA-DRIANOC AE'.

Rev.: Apollo stg. at left, tripod and snake in front. Around: 'CIFA VG-CAESAR'.

Ref.: Kadman 1957, 102 and no. 29.



Site No. 251:

Khirbet Mufyeh

1. Roman, city coin. Bronze, 14.3-15 mm., 5.08 g. worn.

Obv.: Ruler to r., worn.

2. Late Roman, Claudius II Gothicus, 268-270 C.E. Silver alloy, 19-21.6 mm, 3.13 g, Antoninianus, Latin inscr.

Obv.: Ruler to r., adorned with radial crown, clothed; around: 'IMPCCCLAVDIVSAVG'. Pearl edge.

Rev.: Heracles stg. To front, naked, club in r. hand, lion's hide in l.; around: 'IVVENTVSAVG'; at bottom: 'Δ'; pearl edge, attributed to Antioch mint, Axis 6.

Ref.: RIC, V:1, 229, no. 213.



3. Constantine I, 'the Great', 307-337 C.E., Bronze, 15.4-16.5 mm, 1.27 g, AE 3/4, Latin inscr, pearl edge on both sides of coin.

Obv.: Ruler to r., adorned with rosette diadem, clothed, armored; around: 'CONSTANTI-NVS MAXAVG'.

Rev.: Two soldiers stg. with one banner between them, each holding spear in outer hand, inner hand on shield. Around: 'GLORI-IAEXERC-ITVS'; at bottom: 'SMNE'

- Nicomedia mint, V, minted between 330-335 C.E., Axis 12.

Ref.: RIC, VII, Type as in 635, no.199.



4. Constantinople. Bronze, 15.8-17.9 mm, 2.46 g, AE/3, Latin inscr., pearl edge on both sides of coin.

Obv.: Portrait of Constantinople facing l., wearing crested helmet adorned by wreath, clothed, balled staff of outstretched spear on left shoulder; around: 'CONSTANTI...IS'.

Rev.: Victoria turning leftward on ship's prow, spear in r. hand, left hand on shield. At bottom: 'SMA-NA' - Antioch I mint, minted between 330-341 C.E., Axis 6.

Ref.: RIC, VI, type as in 693, nos. 92, 114; Hill and Kent 1965, pl. II:7 or 7:II.



5. Constantine II, 337-340. Bronze, 12.3-13.1 mm, 1.45 g, AE/4, Latin inscr.

Obv.: Ruler to r., adorned with

rosette diadem, clothed, armored.
Around: '...TANTINVS AVG'.

Rev.: Caesar stg. to front, spear upside-down in r. hand, l. hand on shield. Around: 'VIRTVS-...'; at bottom: wreath with R, Rome mint, Axis 12.

Ref.: Bruck 85.



6. Constantius II, 337-361. Bronze, 13.7-15.5 mm, 1.91 g, AE 3/4, Latin inscr.

Obv.: Ruler to r., adorned with pearl diadem; around: 'DNCONSTANTIVS PFAVG'.

Rev.: Two soldiers stg either side of two standards. Around: 'GLORIA EXERCITVS'; at bottom: 'SMAN' - Antioch mint, IV, pearl edge, Axis 6.

Ref.: Bruck 30.



7. Bronze, 13.5 mm, 1.63 g, AE 3/4, indistinct.

Obv.: Ruler to r., adorned by diadem.

Rev.: Victoria to l., Axis 12, third quarter of 4th Century.

Ref.: Bruck 66.

8. Bronze, max. 12.7 mm, 0.72 g, AE 3/4, incomplete, worn.

9. Valentinianus II, 375-392 C.E., Bronze, 11.4-12 mm, 1.04 g, AE/4, Latin inscr.

Obv.: Ruler to r., adorned by diadem. Around: '...NVALENTINIA'.

Rev.: 'XX/MVLT/X/VOT' within wreath; Axis 7; minted between 378-383 C.E.

Ref.: Bruck 93.

10. Theodosius I, 379-395 C.E., Bronze, 11.4 mm, 1.16 g, AE/4, Latin inscription, indistinct.

Obv.: Ruler to r., adorned by diadem, clothed; remnant of inscription: '...OD...'.

Rev.: Victoria dragging captive to l., booty on her shoulder; remnant of inscription: '...PVB'; at bottom: 'CON' - Constantinople mint, Axis 6.

Ref.: Bruck 61.

11. Bronze, 12 mm, 1.08 g, AE/4, illegible.

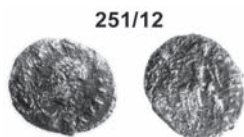
12. Arcadius, 383-408 C.E., Bronze, 12.1-14.2 mm, 1.01 g, AE/4, Latin inscriptions.

Obv.: Ruler to r., adorned with pearl diadem, clothed; around: 'DNARCADIVS PFAVG'.

Rev.: Victoria dragging captive l., booty on her shoulder; Christogram at l.; around: '...VSREI-PUBLICAE'; at bottom: '..SΔ' - Thessalonica mint, IV, minted between 388-393 C.E., Axis 5.

Ref.: RIC, IX, type as in 188, no.

65(c).



13. Honorius, 393-423 C.E., Bronze, 11.7-14 mm, 2.05 g, AE 3/4, indistinct.

Obv.: Ruler to r.

Rev.: Soldier stg at left, spear in r. hand, l. hand on shield, being crowned by Victoria, stg at r. Axis 12.

Ref.: Bruck 85; Sear 1974, type as in 347, no. 4157.

14. Theodosius II, 402-450 C.E., Bronze, 9.4 mm, 0.56 g, Minima, Latin inscriptions, indistinct.

Obv.: Ruler to r., adorned by diadem; around: '...OSIVSPF...'

Rev.: Cross; around" 'CONC...'; Axis 12.

Ref.: Bruck 7.

15. Bronze, 7.1 mm, 0.48 g, Minima, worn. 5th Century C.E.

16. Byzantine, Justinianus I, 527-565 C.E., Bronze, 8.2-9 mm, 0.49 g, Nummia, illegible.

Obv.: Ruler to r., clothed, armored.

Rev.: Large 'A'; Axis 9.

Ref.: Sear 1974a, no. 173, 281.

17. Early Moslem, Umayyad, post-reform of Abd el-Malk, 696-751 C.E., Bronze, 18.2 mm, 2.89 g, anonymous Fils, pomegranate specimen, Arabic inscription, par-

tially broken.

Obv.: In center of first part of Shuhadah: 'LA ALLAH ILA ALLAH / WAHDA'. In margin, around, are two concentric circles, between which three ring-ties are visible.

Rev.: Pomegranate in center, around: second part of Shuhadah: 'MUHAMAD / RASUL / AL...'. In margin, around, are two concentric circles, between which three rings are visible; Axis 8.

Ref.: Walker, type as in 203, no 595, Table XXIII.

18. Bronze, 17.5-19 mm, 0.84 g, Arabic inscriptions, partially broken.

Obv.: In center of first part of Shuhadah: '...LAH / ILA ALLAH / WAHDAH'. In margin, around, are two concentric circles, between which a crescent and point to left of inscription are visible.

Rev.: In center: second part of Shuhadah: 'RASUL A...'. In margin, around, are two concentric circles, between which is name of minter: 'DHARB BIHALB?', to right of inscription. Aleppo mint? Axis 4.

Ref.: Walker, type as in 243, no 794, Table XXVI.

19. Bronze, 14 mm, 1.78 g, Fils, Arabic inscriptions.

Obv.: First part of Shuhadah: 'ALAH / ILA ALLAH / WAHDAH'.

Rev.: 'DHARB HADHA ALFILS BIDIMASHQ', Damascus mint.

Axis 1.

No exact parallel exists. Generally:
Walker 248, nos. 812-815.



20. Bronze, 17-19.2 mm, 4.84 g.
Fils, Arabic inscriptions.

Obv.: 'LA ALLAH ILA / ALLAH
WAHDAH / LA SHARIQ...';
around, remnants of illegible, mar-
ginal inscription.

Rev.: 'ALLAH AHAD AL.../ AL-
SAMD LAM YALD / ...YULD...';
around, marginal inscription: date
and mint (if one exists) outside of
token. Axis 8. Minted after 100 Hi-
jah, 719 C.E.

Ref.: Walker, type as in 242, no.
B.44, pl. XXVI; 291, nos. 954-956,
pl. XXXI.

21. Bronze, 10.2-11.4 mm, 1.06 g,
undefined (coin fragment).

22. Bronze, 11.4-23.5 mm, 1.32 g,
piece of perforated metal sheet.



Site No. 259:

Khirbet Gharur

1. Roman, city coin. Bronze, 17.5-
21 mm, 6.36 g, poor state of pres-
ervation, indistinct. Obverse: Ruler
to r..

2. Bronze, 21 mm, 4.18 g, poor

state of preservation, indistinct.
Obv. Ruler to r.

3. Bronze, max. 22 mm, 3.38 g,
piece of bronze, not definitely coin,
minted depression.

INDEX 1

LIST OF WATER SOURCES IN THE
TERRITORY OF VOL. II
(numbers are those in the text)

No.	Name	Israel Grid	Type
35	'Ain Ginai	1783/2060	Spring
40	Bir el-Haffireh	1726/2009	Spring
49	'Ain Jerbah	1742/1992	Spring
83	'Ain el-Beidan	1801/1853	Spring
85	'Ain el-Far'ah	1829/1885	Spring
86	'Ain el-Miyiteh	1940/1928	Spring
87	'Ain Malih	1940/1928	Spring
88	Hilu Springs	1951/1925	Spring
89	'Ain esh-Shaqq	1972/1946	Spring
91	'Ain Bardaleh	1953/1995	Spring
101	'Ain Miski	1870/1829	Spring
103	Wadi Malih	—	River
104	Wadi el-Far'ah	—	River
106	'Ain Za'anuni	1863/1805	Spring
107	'Ain Jwar	1847/1815	Spring
108	'Ain Mughur 'Id	1862/1809	Spring
109	'Ain Farr	1850/1829	Spring
111	'Ain ed-Dabbur	1875/1803	Spring
112	Jordan River	—	River
115	'Ain Kardaleh	1962/1983	Spring
116	'Ain el-Hammeh	1972/1976	Spring
117	'Ain el-Khimyar	1975/1947	Spring
132	Qa'un Springs	1941/2014	Spring
134	'Ain ej-Jamal	1973/1927	Spring